RESTRUCTURING VS. BANKRUPTCY*

Jason Roderick Donaldson[†] Edward R. Morrison[‡] Giorgia Piacentino[§]

Xiaobo Yu[¶]

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Abstract

We develop a model of a firm in financial distress that can resolve it either by filing for bankruptcy (which is costly) or by restructuring (which is impeded by creditor coordination). We show that bankruptcy and restructuring are *complements*: Lowering bankruptcy costs or making bankruptcy more debtor-friendly can facilitate restructuring rather than crowd it out, which we confirm under a sufficient-statistics condition likely to hold under current U.S. law. The model offers new perspectives on relief policies (e.g., subsidies to bankrupt firms) and on legal debates (e.g., liability management exercises, such as uptiers and dropdowns).

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[†]USC and CEPR.

[‡]Columbia University and NBER.

[§]USC, CEPR, ECGI, and NBER.

[¶]University of Colorado.

1 Introduction

During crises, government policies often target distressed firms in order to avert liquidation and its associated costs, such as supply-chain and labor-market disruptions, thereby mitigating recessions. But such policy interventions help only when private solutions fail. When a firm enters financial distress, two main avenues can avert liquidation: bankruptcy reorganization and out-of-court restructuring. These avenues are about equally common, each accounting for about forty percent of corporate defaults during the past decade (Moody's (2020)).[1,2] Both reduce leverage by exchanging existing debt for new claims against the firm (debt or equity). Restructurings avoid the costs of bankruptcy, such as legal fees and court delays, but are inhibited by a collective-action or "hold-out" problem among dispersed creditors (Bernardo and Talley (1996); Gertner and Scharfstein (1991)).[3] In practice, firms often use a "distressed exchange" of old debt for higher-priority new debt to circumvent this problem.[4] Yet even successful restructurings may not prevent a subsequent bankruptcy: Over one-third are followed by a filing within the next three years (Altman and Kuehne (2020)).[5]

Although restructuring and bankruptcy are individually well understood, much of the literature conflates them or treats them as substitutes.[6] We instead study a distressed firm that may

[1]The rest are missed coupon payments. Recently, out-of-court workouts have accounted for over 60% of defaults (Moody's (2025)).

[2]Similarly, Gilson, John, and Lang (1990) find that about half of financially distressed firms between 1978–87 resolved distress through out-of-court restructuring.

[3]See, e.g., Latham & Watkins Capital Market Practice (2020), Moody's (2017), and Antonoff (2013) for discussions of the holdout problem as a major obstacle to distressed exchanges.

[4]For example, AMC Entertainment, Envision Healthcare, J. Crew, Serta Simmons, and FXI Holdings recently exchanged junior bonds for senior bonds. For a list of recent distressed exchanges, see S&P Global (2025). In several of these restructurings, firms entered well-publicized battles with hold-out creditors. See, e.g., "AMC Entertainment Restructures Debt, Easing Bankruptcy Worries"(Nasdaq.com, August 3, 2020), "J. Crew Holdouts Stumble in Debt-Exchange Lawsuit" (*Wall Street Journal*, April 26, 2018), "Fitch Rates SM Energy 'CCC+' on Completion of Debt Exchange," (*Fitch Ratings*, June 22, 2020), "Envision Healthcare Announces Completion of Debt Exchange Transactions," (*Bloomberg*, May 1, 2020), "AMC Theaters Considers Bankruptcy After Moviegoers Stay Home," (*Bloomberg*, October 13, 2020), and "Supreme Court Declines to Hear Serta Simmons Lender Dispute," (*Wall Street Journal*, Nov. 10, 2025). See James (1995) for historical evidence on the importance of priority in resolving holdout problems among bondholders during restructurings.

[5]Among the examples cited in the preceding footnote, J. Crew and Serta Simmons ultimately filed for bankruptcy; see Buccola and Nini (2024) and Ayotte and Scully (2021).

[6]See, e.g., Asquith, Gertner, and Scharfstein (1994), Becker and Josephson (2016), Favara, Schroth, and Valta (2012), Franks and Torous (1994), Gertner and Scharfstein (1991), and Gilson, John, and Lang (1990). In some

file even after a successful restructuring. We ask: How do key bankruptcy parameters—its deadweight costs and "creditor friendliness"—affect the likelihood of an out-of-court restructuring? Which policies best mitigate corporate distress, and which can boost credit supply? Are subsidies to bankruptcy,[7] which could target the most distressed firms, optimal? Or can subsidies to restructuring,[8] which could prevent distress in the first place, do better?

Our model challenges conventional wisdom. We find that an efficient bankruptcy system encourages out-of-court restructurings—bankruptcy and restructuring are complements, not substitutes. A more creditor-friendly system can impede restructuring, depending on the characteristics of the bankruptcy regime. Applying a sufficient-statistics approach to estimates for the U.S., we find that, indeed, making the Code more creditor-friendly would likely make restructuring harder here.

Our policy analysis suggests that subsidies to bankrupt firms can backfire by raising bankruptcy costs and constraining credit supply, whereas subsidies to secured creditors—including those of bankrupt firms—can be socially optimal, a conclusion that, remarkably, holds regardless of whether the planner's goal is to minimize ex-post bankruptcy costs or to maximize ex-ante debt capacity.

Model preview. A distressed firm has risky assets v and debt D_0 held by dispersed creditors. There are two dates. At date 0, before v is realized, the firm can propose a restructuring of its debt. At date 1, v is realized, and the firm has a choice: repay the debt or file for bankruptcy.[9] Bankruptcy destroys $(1-\lambda)v$ of value, and creditors bargain with equity over the remaining λv, of which creditors capture fraction θ—our measure of "creditor friendliness," parameterizing the legal environment. Thus, a restructuring that reduces D_0 decreases expected deadweight loss and, as such, can make everyone better off.

papers, such as Fan and Sundaresan (2000) and Hart and Moore (1994, 1998), bankruptcy serves as the outside option for renegotiation. These papers, however, typically do not model the bankruptcy choice, which is instead synonymous with liquidation.

[7]See, e.g., DeMarzo, Krishnamurthy, and Rauh (2020).

[8]See, e.g., Blanchard, Philippon, and Pisani-Ferry (2020); Greenwood and Thesmar (2020).

[9]It could also file for bankruptcy at date 0 (but we show that this is never optimal) or propose a restructuring at date 1 (but we show that this will never be accepted by creditors).

Two assumptions are key to our results. The first is that when each creditor decides whether to accept a restructuring offer, it takes the decisions of other creditors as given. This is what we mean when we say creditors are "dispersed": They cannot act collectively. The second is that the firm chooses whether to file for bankruptcy, and can act strategically: It can choose to file in order to benefit from a debtor-friendly code ($\theta < 1$) even if it could repay its debt in full. Both assumptions align with practice. Regarding the first: Corporate debt is often dispersed: Bonds accounted for over half of all corporate debt issuances during 2020 (He and Sun (2021)), and the median bond issuance had 86 institutional investors (Li and Haiyue (2023)). Even syndicated debt is dispersed: Although the median has only 11 investors, the average syndicate has over 100 members (Siedlarek and Yankov (2025)). And dispersed creditors are well-understood to hinder restructurings (Morrison (2009)). Regarding the second assumption: 98% of bankruptcies are voluntary filings (Hynes and Walt (2020)), including well-known "strategic" cases (Cole (2002); Moody's (2006)). Indeed, the median corporation filing for bankruptcy is solvent, with a leverage ratio below 1 (Eckbo et al. (2023), Seth (2025)).

Results preview. In our model, the hold-out problem dooms any restructuring involving an exchange of debt for equity or for equal-priority debt: An individual creditor knows that if others accept the offer, the firm will avoid distress and likely be able to pay its debt in full. Because creditors are dispersed, each has an incentive to hold out. However, as in Bernardo and Talley (1996) and Gertner and Scharfstein (1991), the firm can restructure if it offers to exchange existing debt for *higher-priority* debt. Creditors accept a write-down in the face value of the debt (which decreases their payment if the firm does not file) in exchange for an increase in their priority (which increases their payment if it does file).

Creditors accept a restructuring if priority is valuable, which it is when (i) bankruptcy is sufficiently likely and (ii) recoveries are meaningful. That is our first insight. It reframes "bargaining in the shadow of the law" (Mnookin and Kornhauser (1979)): Bankruptcy serves as an *inside* option (relevant if restructuring succeeds) rather than an outside option (relevant only if it fails).

Our analysis hinges on a second insight as well: The probability of bankruptcy, and hence the value of priority, is determined by a strategic decision of the firm, namely, whether to file. As is typical in the hold-out literature—in which bankruptcy is generally *not* a strategic decision—the firm in our model files for bankruptcy when its asset value v falls below a threshold, which we denote by $\hat{v}$. But, unlike in that literature, $\hat{v}$ depends on the parameters of the bankruptcy environment: Bankruptcy is more attractive to the firm when bankruptcy costs are low (λ is high) and when the Code is debtor-friendly (θ is low). Hence, the bankruptcy filing threshold, $\hat{v}$, is increasing in λ and decreasing in θ, all else equal. But all else is not equal, as these parameters also have an effect on creditors' decision to accept a restructuring offer—an effect that turns out not to be obvious.

This leads to our first main result: A decline in bankruptcy costs (an increase in λ) facilitates restructuring. To see why, recall that restructuring is feasible only insofar as creditors are willing to accept write-downs in exchange for priority. And recall that the value of priority has two components: (i) the probability of bankruptcy, and (ii) the recovery value in bankruptcy. Both increase as λ increases: An increase in λ makes it more attractive to the firm to file (an indirect effect) and increases recovery values for senior debt (a direct effect).

Our second main result is that excessively creditor-friendly bankruptcy law (high θ) can deter out-of-court restructuring. As with λ, the write-down-maximizing θ maximizes the value of priority, which has the same two components. But now they push in opposite directions. Increasing θ has a positive effect on the value of priority because it increases recovery values for senior debt, but it also has a negative effect because it makes filing for bankruptcy less attractive to the firm as more value is diverted to creditors. Whether increasing θ helps or hinders restructuring depends on which effect is stronger.

We derive a condition to test whether creditor friendliness crowds out restructuring. Using sufficient statistics, we show this condition is likely satisfied in the U.S. today. This implies that restructurings, which avoid the deadweight costs of bankruptcy, would be more prevalent in the U.S. if bankruptcy laws were made *less* creditor-friendly. This finding is consistent with practitioners' intuition (Miller and Marcus (1989)).

We then use our model to evaluate macro policy during a crisis. Governments often respond to crises by subsidizing corporations in order to prevent default and bankruptcy. Examples in the U.S. include subsidies to small and large corporations during the COVID pandemic (e.g., the CARES Act and the Fed's Primary Market Corporate Credit Facility), capital investments made by the Treasury during the Great Financial Crisis (via the Troubled Asset Relief Program), and loans to individual corporations during recessions (e.g., the Chrysler Corporation Loan Guarantee Act of 1979).[10]

Our model provides a framework for thinking about optimal government policy in situations like these. We analyze how a planner should allocate a marginal dollar among subsidies to each class of claim—equity, secured debt, and unsecured debt—inside or outside bankruptcy. We first consider a planner who aims to maximize social welfare ex interim (after firms have incurred debt). Our third main result is that it does not matter whether you subsidize (i) secured debt (in or out of bankruptcy) or (ii) equity out of bankruptcy: The planner can incentivize (i) creditors to reduce debt in restructuring (by increasing the value of priority) or (ii) the firm not to file (by paying when it does not). Both are optimal. Other subsidies can backfire. Subsidies to equity in bankruptcy induce excessive filing, and those to unsecured creditors create holdouts.

We also consider a planner seeking to maximize debt capacity ex ante. Our fourth main result is that, remarkably, the optimal policies are robust to the planner's objective: The same policies that maximize welfare ex interim—subsidies to secured debt and to equity out of bankruptcy—also maximize debt capacity ex ante. Intuitively, paying equity outside of bankruptcy both encourages repayment (increasing debt capacity) and avoids distress costs (increasing welfare). That is equivalent to subsidizing secured debt, which does the same thing by facilitating restructuring.

Together, our results speak to familiar policy debates. For instance, the finding that subsidies

[10] These policy interventions have been widely discussed, including by Casey (2021), Levitin (2011), and Reich (1985).

in bankruptcy should go to secured debt supports allocating value by seniority. Such subsidies ultimately benefit equity, as they foster borrowing ex ante and restructuring ex interim. Likewise, we show that pandemic-era grants and forgivable loans (favoring unsecured debt) may hinder restructuring,[11] whereas measures that facilitate restructuring directly or aid secured creditors are more effective.[12]

Extensions. Section 6 and Appendix D develop several extensions. (i) We analyze liability management exercises (LMEs)—known as "uptiers" and "dropdowns"—that can create super-senior debt, possibly stripping existing secured creditors of their status at the top of the priority hierarchy. These LME's are often viewed as manifestations of "creditor-on-creditor violence," as discussed by Buccola and Nini (2024) and Antill et al. (2025). We present two irrelevance results: Being anticipated, the possibility of the LME does not affect the extent of restructuring or allocation of surplus among creditors. Neither does the legal status of supersenior debt. (ii) We allow creditors to be concentrated as well as dispersed. With concentrated creditors, restructurings may include debt-for-equity swaps; with dispersed creditors, only debt-for-debt swaps occur. (iii) We study court congestion. We show that bankruptcy policy can amplify crises and thus matters for financial stability. (iv) We account for debt overhang and show that, although such costs always increase the benefits of restructuring, their effect on the likelihood of restructuring is ambiguous. (v) We endogenize borrowing and show that our results are robust. (vi) Finally, when we account for tort claimants and other involuntary creditors, optimal priority places them behind secured but ahead of unsecured creditors.

Literature. Our paper bridges two strands of the bankruptcy literature. One focuses on the hold-out problem as an impediment to restructuring.[13] Roe (1987) was among the first to focus

[11]These include, e.g., the loan guarantees proposed by Blanchard, Philippon, and Pisani-Ferry (2020) and policies such as the Payroll Support Program (PSP) and Paycheck Protection Program (PPP). United Airlines, for example, was offered \$5 billion of assistance through the PSP. The \$5 billion included a \$3.5 billion grant and a \$1.5 billion low-interest rate loan, as reported in United Airlines (2020)

[12]Examples include restructuring-support programs (Blanchard, Philippon, and Pisani-Ferry (2020); Greenwood and Thesmar (2020)) and tax-based incentives, such as IRS Regulation TD9599 (2012), which reduced taxes on creditor losses and doubled restructuring rates (Campello, Ladika, and Matta (2018)).

[13]Our paper complements papers studying other restructuring frictions, such as asymmetric information (Bulow and Shoven (1978), Giammarino (1989), and White (1980, 1983)). Our work departs from papers in which such

on this problem in the context of bondholders, whose inability to coordinate (exacerbated by federal law) can prevent efficient restructuring and render bankruptcy necessary.[14] Gertner and Scharfstein (1991) study the problem more formally, showing that a debtor can induce claimants to agree to a restructuring via an "exchange offer" that offers seniority to consenting creditors (and thereby demotes non-consenting creditors).[15] Bernardo and Talley (1996) show that the ability to make such exchange offers can distort a firm's investment incentives.[16] In these papers, however, bankruptcy is not a choice; it is an automatic consequence of the firm's inability to pay its debts.

A separate strand of the literature focuses on the bankruptcy decision and explores the effects of bankruptcy rules, such as the APR, on this decision. Baird (1991) and Picker (1992), for example, assess whether these rules induce firms to enter Chapter 11 when doing so maximizes recoveries to dispersed unsecured creditors. Picker (1992) concludes that, because the filing decision is held by shareholders, optimal rules might permit violations of the APR in order to induce filings that maximize ex post recoveries. These papers, however, do not consider how rules affecting the bankruptcy filing decision also affect the likelihood of a successful restructuring ex ante.[17]

Our paper is also related to several other lines of research. A large literature studies the effects of creditor priority on bankruptcy outcomes and ex ante investment decisions (examples include Adler (1995) and Bebchuk (2002)). Recent work has focused on the optimal "creditor friendliness" of bankruptcy laws, showing that the optimal level depends on judicial ability in

frictions are absent and, as a result, Coasean bargaining among investors leads to efficiency (e.g., Baird (1986), Haugen and Senbet (1978), Jensen (1986), and Roe (1983)).

[14]In corporate finance, this idea is also central to Grossman and Hart's (1980) model in which free-riding shareholders refuse efficient takeovers.

[15]Roe and Tung (2016) also study exchange offers and show that a successful exchange can nonetheless be followed by a bankruptcy filing.

[16]Haugen and Senbet (1988) discuss ways to solve the coordination problem contractually (though some of the solutions could run afoul of the Trust Indenture Act). For example, the indenture could permit the firm to repurchase the bonds at any time at a specified price (e.g., the price quoted in the most recent trade).

[17]Another strand of the literature is exemplified by Mooradian (1994), Povel (1999), and White (1994), who view bankruptcy as a screening device that can induce liquidation of inefficient firms and the reorganization or restructuring of efficient firms.

bankruptcy and the quality of contract enforcement outside of bankruptcy (see Ayotte and Yun (2009)) as well as on the extent to which default imposes personal costs on owners and managers (see Schoenherr and Starmans (2020)).[18] Our work contributes to this literature because we show how creditor friendliness in bankruptcy (ex post) affects the restructuring decision ex ante.

Our paper also contributes to research on the determinants of debt structure (recently surveyed by Colla, Ippolito, and Li (2020)) and the drivers of debt renegotiation (e.g., Roberts and Sufi (2009)).

Layout. Section 2 presents the model. Section 3 characterizes the equilibrium and shows that restructuring is feasible only when priority is valuable. Section 4 derives our main results on bankruptcy efficiency and creditor friendliness. Section 5 studies policy interventions. Section 6 presents extensions. Section 7 concludes. Proofs and derivations appear in the Appendix.

2 Model

We study a two-date model of a single firm with (positive) asset value $v \sim F$ and initial debt D_0 owed to a unit continuum of identical, risk-neutral creditors.[19] The firm is controlled by risk-neutral equity holders (or managers/directors acting in their interest).

2.1 Restructuring

At date 0, the firm can avoid distress, making a take-it-or-leave-it exchange offer to creditors to reduce debt to $D < D_0$. Being dispersed, creditors each decide whether to accept the offer taking others' decisions as given.[20] We focus on equity and debt, the most common claims in real-world restructurings (Gilson, John, and Lang (1990)), but discuss more general claims in

[18] Sautner and Vladimirov (2017) also study optimal creditor friendliness, showing that greater creditor friendliness can facilitate ex ante restructuring when the firm has a single creditor who is unsure about firm cash flows during restructuring but sure about them in bankruptcy.

[19] See Appendix D.3 on the determination of D_0.

[20] We assume restructuring takes the form of an exchange offer, as is typical for U.S. corporate bonds subject to the Trust Indenture Act (TIA), which prohibits changes to core bond terms without unanimous consent (see Hart (1995), Ch. 5). Similar clauses appear in syndicated loans (Sufi (2007)).

Appendix E.[21]

2.2 Financial Distress: Bankruptcy

In our model, D denotes the firm's debt at the end of date 1—the outcome of a restructuring if one occurs, and D_0 otherwise. If the firm repays D in full, creditors receive D and equity gets $v - D$. Otherwise, the firm can file for bankruptcy, incurring deadweight costs $(1 - \lambda)v$, reflecting legal fees, delays, and operational disruptions (e.g., Titman (1984)).[22] Usually bankruptcy destroys value ($\lambda < 1$), but sometimes it can do the opposite; for example, it could enable asset sales "free and clear" of liabilities or halt foreclosures, a case captured by allowing $\lambda > 1$. Although we stress that case only in Section 5, all of our results apply to it unless stated otherwise.

If the firm files, investors recover λv, of which creditors capture a fraction θ. θ measures the system's "creditor friendliness" or creditors' bargaining power (as modeled explicitly in Appendix A). If $\theta < 1$, equity retains some value even when creditors are not paid in full.[23]

The resulting payoffs are:

$$\text{Equity payoff } = \begin{cases} v - D & \text{if repayment,} \\ \\ (1-\theta)\lambda v & \text{if bankruptcy.} \end{cases} \tag{1}$$

In practice, however, some offers require a minimum acceptance rate. These provisions make no difference to our baseline analysis with a continuum of creditors, but they could with a finite number of creditors (Bagnoli and Lipman (1988); Section 6.2).

[21]We abstract from covenants that could restrict new senior debt issuance, such as "negative pledge" clauses. These can typically be removed through exit consents with majority approval (Kahan and Tuckman (1993)) and offer weak protection against dilution (Bjerre (1999); Donaldson, Gromb, and Piacentino (2020a)).

[22]We focus on ex post costs in the baseline model and include ex ante costs in Appendix D.2. See, e.g., Davydenko, Strebulaev, and Zhao (2012) and Dou et al. (2020) for estimates of such costs. These costs can also capture the rise of devices to speed up the bankruptcy process, such as the so-called prepackaged bankruptcies that allow the firm to obtain creditor consent to its plan before a bankruptcy case is filed.

[23]Equity may benefit through explicit APR deviations or indirect mechanisms such as restructuring support agreements, third-party releases, or the appointments of bankruptcy directors to stymie suits against shareholders. Earlier studies (e.g., Eberhart, Moore, and Roenfeldt (1990); Weiss (1990)) document frequent APR violations before 2000; these are now rarer (Wang (2022)). See also Buccola (2022), Ellias, Kamar, and Kastiel (2022), and Levitin (2022). We revisit the efficiency of this shift in Section 4.3.

$$\text{Debt payoff } = \begin{cases} D & \text{if repayment,} \\ \\ \lambda\theta v & \text{if bankruptcy.}^{24} \end{cases} \tag{2}$$

Thus, the key parameters—bankruptcy efficiency (λ) and creditor friendliness (θ)—capture the difference between restructuring and bankruptcy.

2.3 Timeline

The sequence of moves is as follows. At date 0, the firm can restructure its debt or leave it unchanged. At date 1, the asset value v is realized; the firm then either repays D or files for bankruptcy.[25]

3 Equilibrium Characterization and the Value of Priority

We use backward induction to characterize the symmetric pure-strategy subgame-perfect equilibrium. Restructuring faces a classic hold-out problem, which can be resolved by exchanging old debt for new, higher-priority debt. Building on this, we show that the feasible write-down in restructuring increases with the value of priority in bankruptcy.

[24] We focus on asset values rather than cash flows because firms with dispersed debt typically face solvency, not liquidity, problems. Liquidity problems (low cash flows) are less likely to cause bankruptcy because such firms can likely raise capital via refinancing, capital injections, or asset sales (see Black and Cox (1976); Leland (1994); Chaney, Sraer, and Thesmar (2012)).

[25] We focus on restructuring at date 0 and bankruptcy at date 1. Allowing both choices at both dates changes nothing: Filing at date 0 is suboptimal (it eliminates the option to recover) and restructuring at date 1 is infeasible (see Appendix B).

3.1 Default and the Bankruptcy Filing Decision

At date 1, given v and D, the firm chooses between repayment and bankruptcy. From equation (1), it files when $(1-\theta)\lambda v > v - D$, i.e. when:

$$v < \hat{v}(D) := \frac{D}{1-(1-\theta)\lambda}. \tag{3}$$

If bankruptcy destroys all value ($\lambda = 0$) or is fully creditor-friendly ($\theta = 1$), the firm files only when $v < D$. Otherwise, the firm may file even when solvent ($v > D$).[26]

3.2 Infeasible Restructuring

One way to avoid bankruptcy is to eliminate debt through an equity conversion:

Lemma 1. *For $\lambda < 1$, restructuring debt into equity worth a fraction $1-\varphi$ of assets makes both firm and creditors strictly better off for any φ in a non-empty interval (given in the proof).*

Yet, dispersed creditors will hold out and reject what is collectively efficient:

Lemma 2. *No ex ante debt-to-equity restructuring exists that both uncoordinated creditors are willing to accept and the firm is willing to offer.*

Each creditor knows that if others accept the restructuring, the firm will be solvent and, therefore, prefers to hold out, knowing its debt will be paid in full.[27] The same logic makes pari passu debt exchanges (equal-priority debt-for-debt swaps[28]) infeasible:

[26] Insolvency is not required for filing under U.S. law. Courts may allow Chapter 11 if a firm's financial health is "threatened" by existing debt, even without immediate default (*In re LTL Management, LLC*, 64 F.4th 84, 102 (3d Cir. 2023)). Indeed, among large corporations, the median bankruptcy filer is (marginally) solvent, as noted above.

[27] The hold-out arises because no creditor is pivotal. Requiring unanimous consent could solve it, but such offers are rarely used—"it takes only one ('crazy') creditor ... to defeat a unanimity offer" (Hart (1995), p. 116).

[28] It may be useful to illustrate how a decrease in debt can make creditors better off: It can be better to receive less with a higher probability than more with a lower one. Creditors with debt D_0 are better off decreasing debt if

$$\left.\frac{\partial}{\partial D}\right|_{D=D_0}\left(\Big(1-F\big(\hat{v}(D)\big)\Big)D + F\big(\hat{v}(D)\big)\mathbb{E}\left[\lambda\theta v \,\middle|\, v < \hat{v}(D)\right]\right) < 0$$

Lemma 3. *No ex ante restructuring of debt to equal-priority debt that creditors are willing to accept and the firm is willing to offer exists.*

3.3 Feasible Restructuring

A restructuring can overcome the hold-out problem by swapping old debt for new, more senior debt. By granting priority in exchange for a write-down, such an offer becomes feasible.[29] This is an old result. We say something new by analyzing the size and drivers of the write-down $D_0 - D$, as we now explore.

A creditor accepts the offer only if the expected value of senior debt (with face value D) exceeds that of junior debt (with face value D_0), given that the restructuring succeeds (so the aggregate face value is also D, which the individual creditor, being small, takes as given). The incentive constraint (IC) is:

$$\Delta := \Big(1 - F(\hat{v}(D))\Big)D + F(\hat{v}(D))\,\mathbb{E}[\lambda\theta v \mid v < \hat{v}(D)] - \Big(1 - F(\hat{v}(D))\Big)D_0 \geq 0. \tag{4}$$

The first two terms of Δ constitute the creditor's expected payoff from accepting the offer: If there is no future bankruptcy, it gets D; if there is one, it gets a unit share of the recovery. The last term is its expected payoff from holding out: If there is no future bankruptcy, it gets D_0; if there is, it gets zero. The reason it gets zero in this case is that the bankrupt firm cannot pay all debt in full (since $v \leq \hat{v}$ in that case, $D > \lambda\theta v$ by equation (3)) and all other debt is senior (by virtue of the restructuring), making the payoff to junior debt zero.[30]

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or, computing,

$$\frac{1 - F\big(\hat{v}(D_0)\big)}{f\big(\hat{v}(D_0)\big)\hat{v}(D_0)} < \frac{1-\lambda}{1-(1-\theta)\lambda}.$$

Two observations: (i) The condition can be satisfied only if λ is sufficiently small: If $\lambda = 1$, there are no bankruptcy costs to avoid by reducing debt, so creditors are always better off with more debt; (ii) it can be satisfied more easily when $f\big(\hat{v}(D_0)\big)$ is large—that is, when a small reduction in debt from D_0 has a significant impact on the probability of default. At any rate, restructurings always increase total surplus in our model, even if they do not implement Pareto improvements.

[29] See, e.g., Diamond (1993), Donaldson, Gromb, and Piacentino (2020b, 2020a), and Donaldson, Piacentino, and Yu (2022).

[30] Here, senior debt is always paid ahead of junior debt—there are no deviations from the APR that favor junior

The firm offers the largest write-down that satisfies the IC, yielding:

Proposition 1. Restructuring to senior debt: *Let D^* denote the equilibrium face value. The maximum write-down all creditors accept is:*

$$D_0 - D^* = \frac{F(\hat{v})}{1 - F(\hat{v})}\, \mathbb{E}[\lambda\theta v \mid v < \hat{v}]. \tag{5}$$

Accepting is a dominant strategy.

The feasible write-down $D_0 - D^*$ increases with (i) the probability of bankruptcy $F(\hat{v})$ and (ii) creditors' recovery in default $\mathbb{E}[\lambda\theta v \mid v \leq \hat{v}]$. These two ingredients, which underlie the rest of our analysis, reflect the value of priority afforded by senior debt: (i) If a firm never goes bankrupt, priority has no value (even the most junior creditor is paid in full) and (ii) if total recovery value is sufficiently low, priority has little value (even senior debt is paid little).

The expression for the write-down depends on the parameters of the bankruptcy environment, λ and θ, both via the recovery in bankruptcy (a direct effect) and via the filing threshold $\hat{v}$ (an indirect effect). "Naïve" comparative statics (which ignore the indirect effect) would capture a benchmark in which bankruptcy is not strategic (e.g., due to "liquidity default"). Such comparative statics would suggest that the write-down is increasing in λ and in θ.[31] We show next that when bankruptcy is a choice made by the firm, these comparative statics can be amplified—or even reversed.

creditors at the expense of senior. Although this assumption appears to be a good approximation of reality (Bris, Welch, and Zhu (2006)), we relax it in Section 6.1.

[31] This naïve comparative static is arguably too naïve, even for a benchmark, because it ignores not only that $\hat{v}$ depends on λ and θ, but also that it depends on D itself. However, the naïve conclusions do not change if $\hat{v}$ is a function of D (alone). To see why, let $G(\hat{v}) := \frac{F(\hat{v})}{1-F(\hat{v})}\mathbb{E}[v|v \leq \hat{v}]$. Thus, $D_0 - D^* = \lambda\theta G(\hat{v})$. Differentiating with respect to λ and rearranging yields: $\frac{\mathrm{d}D^*}{\mathrm{d}\lambda} = -\frac{\theta G(\hat{v})}{1+\lambda\theta G'(\hat{v})\frac{\mathrm{d}\hat{v}}{\mathrm{d}D}}$. Given that $G' > 0$, this is negative as long as bankruptcy is more likely when debt is higher, ceteris paribus, $\partial\hat{v}/\partial D > 0$, an assumption that holds in all models we are aware of. The analysis for θ is identical.

4 Analysis of the Bankruptcy Environment

We now study how the bankruptcy environment affects restructuring, focusing on the key parameters λ (efficiency) and θ (creditor friendliness). Both influence restructuring directly (via recoveries) and indirectly (via the firm's bankruptcy decision). We then do two quantitative exercises using estimates from the literature: (i) We validate our predictions, comparing our model-predicted estimates to observed write-downs, and (ii) we derive a sufficient statistics condition, using it to argue that a reduction in creditor friendliness would likely facilitate restructuring in the U.S. today.

4.1 Bankruptcy Costs and Restructuring

Differentiating the face value D^* with respect to λ, we obtain the next result:

Proposition 2. BANKRUPTCY COSTS: *Reducing bankruptcy costs (increasing λ) facilitates restructuring in the sense that the maximum write-down $D_0 - D^*$ is increasing in λ.*

This is a central result of our paper: Restructuring and bankruptcy are complements, not substitutes. This is true for two reasons, which echo the ingredients that make priority valuable (see Proposition 1):

(i) The more efficient bankruptcy is, the more likely the firm is to file, and priority in bankruptcy is more valuable when it is more likely.

(ii) The more efficient bankruptcy is, the more creditors recover in bankruptcy, and priority in bankruptcy is more valuable when recovery values are higher.[32]

In other words, a high value of priority, per (i) and (ii), gives the firm something valuable to offer in exchange for the write-down, facilitating restructuring.

[32] To see why, recall, from the IC in inequality (4), that creditors accept a restructuring only if the payoff in bankruptcy from accepting senior debt is high relative to the payoff in bankruptcy from holding out, which, conditional on others accepting, is equal to zero. Thus, as senior creditors' payoff in bankruptcy increases, so does the write-down creditors are willing to accept.

4.2 Creditor Friendliness and Restructuring

The feasibility of a restructuring also depends on the creditor friendliness of the bankruptcy system. Differentiating the equilibrium face value D^* with respect to θ gives:

Proposition 3. Creditor friendliness*: An increase in creditor friendliness (θ) facilitates restructuring—i.e. increases the maximum write-down $D_0 - D^*$—iff $\partial\Delta/\partial\theta > 0$ (see (61) in the Appendix). Moreover, if*

$$\frac{1 - F(D^*_{\theta=1})}{D^*_{\theta=1} f(D^*_{\theta=1})} < \lambda, \tag{6}$$

where $D^*_{\theta=1}$ solves $\Delta = 0$ at $\theta = 1$, then the write-down is maximized at an interior $\theta^* \in (0, 1)$.

The ambiguity in the result arises because creditor friendliness affects the value of priority in two opposing ways (cf. Proposition 1):

(i) By increasing creditors' recovery in bankruptcy, creditor friendliness makes priority *more* valuable.

(ii) By reducing equity's payoff in bankruptcy, creditor friendliness lowers the firm's incentive to file, making priority *less* valuable.

Condition (6) implies that when θ is already high, the second effect dominates: Further increases in θ reduce the feasible write-down. Hence, the write-down maximizing level satisfies $\theta^* < 1$.[33]

This result hinges on the assumption that firms choose whether to file for bankruptcy. If bankruptcy were not strategic—as is typical in the hold-out literature—the filing probability would not depend on θ. Then raising θ would only increase recoveries and, per (5), would always facilitate restructuring. A caveat is that we have taken D_0 as exogenous. One might worry about a Lucas-type critique: The bankruptcy environment also affects the amount of debt issued in the first place. But our results are robust to endogenizing D_0, as we show in Appendix D.3.

[33] Bisin and Rampini (2005) identify a related downside of creditor friendliness: Lower payoffs to equity can hinder enforcement of exclusive contracts.

4.3 Application to U.S. Law: Model Validation and Sufficient Statistics for θ to Crowd Out Restructuring

Here we validate the model using U.S. data on write-downs and credit spreads. Then we use it to assess whether existing bankruptcy law crowds out restructuring in the U.S. today.

We begin by expressing the write-down in terms of observable yields. Rewriting inequality (5) gives $De^{-y^s} = D_0 e^{-y^u}$, where y^s is the yield on senior (secured) debt (received in a restructuring) and y^u is the yield on junior (unsecured) bonds (which are exchanged), conditional on the write-down. Rearranging and approximating gives:[34]

$$\text{\% write-down} \;\approx\; \text{secured credit spread.} \tag{7}$$

Thus, creditors accept write-downs only to the extent that seniority is valuable (as measured by the secured credit spread).

Using estimates from Benmelech, Kumar, and Rajan (2020), distressed firms exhibit an unannualized secured–unsecured spread of about 42 percent (about six percent annualized for bonds with maturity of about seven years), consistent with the mean 44 percent write-down documented in Mooradian and Ryan (2005) for distressed exchanges of unsecured for secured debt.

Having verified that the model aligns with observed spreads and write-downs, we turn to whether increasing creditor friendliness under U.S. law would crowd out restructuring. We impose two mild assumptions: (i) D^* is a continuous function of θ with a unique local minimum[35] and (ii) $vf(v)$ is increasing on $[0, \hat{v}]$.[36] In Appendix G.1 we show that a sufficient condition for an increase in θ to crowd out restructuring is

$$\frac{D^*}{D_0} \;\leq\; \frac{1 - \lambda(1-\theta)}{2 - \lambda(2-\theta)}. \tag{8}$$

[34] From the binding IC, we have $De^{-y^s} = D_0 e^{-y^u}$ or, rearranging, $y^u - y^s = \log(D_0/D) = -\log(1 - (D_0 - D)/D_0)$. Applying the approximation $\log(1-x) \approx -x$ to the last term gives the result.

[35] This holds for commonly used distributions such as the uniform; see Appendix C.

[36] This is satisfied, for example, if f is unimodal and $\hat{v}$ lies below its mode.

This condition captures the two competing effects of greater creditor friendliness on the value of priority: (i) higher recovery values in bankruptcy (which raise the value of priority) and (ii) lower filing incentives for equity (which reduce the likelihood that priority matters). The right-hand side is increasing in θ, so effect (i) dominates when θ is low, whereas effect (ii) dominates when θ is high.

Using conservative estimates given in Appendix G.2 ($\lambda = 0.9$, $\theta = 0.85$, and $D^*/D_0 = 0.44$), condition (8) requires

$$56\% \;\leq\; \frac{1 - 0.9(1 - 0.85)}{2 - 0.9(2 - 0.85)} \approx 90\%,$$

which is satisfied by a wide margin. Hence, we conclude that a marginal reduction in creditor friendliness would likely increase the feasibility of restructuring in the U.S. today.[37]

Of course, these estimates are averages; parameters vary across firms. We therefore examine robustness. First, we show that condition (8) holds for a wide range of λ and θ (Figure 1). Second, using the explicit solution for D^* under a uniform F (Appendix C), we verify that the necessary-and-sufficient condition in Proposition 3 is satisfied for a broad region of the parameter space (Figure 2). These conclusions are insensitive to the specific uniform parameters. Taken together, both approaches indicate that under current U.S. law, decreases in creditor friendliness would likely facilitate restructuring.[38]

5 Policy Analysis

We now turn to the policy implications of our model. We take the perspective of a social planner allocating a marginal dollar (a "subsidy") across different parts of the capital structure. We allow the planner to pursue one of two objectives. The first is to minimize the costs of bankruptcy—a natural objective in crisis times, when debt is already in place, and the goal is to limit deadweight

[37] Dávila (2020) also uses a sufficient statistics approach to analyze bankruptcy policy, albeit in the context of household finance. His analysis points to a distinct benefit of debtor friendliness ("exemptions"): insurance against income risk.

[38] This mechanism may also help explain the rise in bankruptcy filings following a pro-creditor Supreme Court decision, as documented in Giambona, Lopez-de Silanes, and Matta (2019).

losses. It coincides with maximizing welfare here, subject to the proviso that D_0 is taken as given. The second is to maximize debt capacity—a natural objective in normal times, when policy aims to support credit supply to productive firms when D_0 is endogenous. We emphasize the case in which bankruptcy is inefficient ($\lambda < 1$), but consider policies that are robust to the contrary ($\lambda > 1$).

After setting up the planner's problem, we study how subsidies to each layer of the capital structure—inside and outside bankruptcy—affect distress. Each policy has both a "direct effect" (how it changes payoffs conditional on distress) and an "indirect effect" (how it changes restructuring incentives). We then apply these results to a set of concrete policies, ranging from long-standing bankruptcy reforms to pandemic-era interventions.

Focusing on subsidies is a useful approximation to real world policy-making, as illustrated by the airline bailouts during COVID-19. Lee (2021) explains that these interventions included unsecured loans (subsidizing debt), equity infusions (subsidizing both equity and debt), and wage and flight subsidies (analogous to blanket asset subsidies).

5.1 Planner's Problem for a Marginal Dollar

Consider a planner with a small budget ε to be allocated across subsidies $\mathbf{s} \geq 0$, with associated expected costs $\mathbf{q}$. The planner has no other tools and must respect creditors' incentive compatibility (IC), given by inequality (4) (equivalently, $\Delta(\mathbf{s}) = 0$). The planner's problem is

$$\begin{cases} \max & \mathscr{U}(\mathbf{s}) \\ \text{s.t.} & \Delta(\mathbf{s}) = 0, \\ & \mathbf{q} \cdot \mathbf{s} = \varepsilon, \end{cases} \tag{9}$$

over feasible subsidies $\mathbf{s}$.

We allow for a general set of subsidies: The planner may subsidize each layer of the capital structure—equity (E), unsecured debt (U), and secured debt (S)—conditional on the firm being either inside bankruptcy (B) or outside it (O). Thus, $\mathbf{s} = (s_E^B, s_U^B, s_S^B, s_E^O, s_U^O, s_S^O)$.

With these policies, the creditors' IC changes. Creditors prefer restructuring only if a senior claim of face value D (plus outside-bankruptcy subsidy s_S^O and inside-bankruptcy subsidy s_S^B) is worth at least as much as the junior claim of face value D_0 (plus s_U^O and s_U^B):

$$\begin{aligned}(1 - F(\hat{v}(\mathbf{s})))\,(D + s_S^O) + F(\hat{v}(\mathbf{s}))\,\mathbb{E}[\lambda\theta v + s_S^B \mid v \leq \hat{v}(\mathbf{s})] \\ \geq (1 - F(\hat{v}(\mathbf{s})))\,(D_0 + s_U^O) + F(\hat{v}(\mathbf{s}))\,s_U^B.\end{aligned} \tag{10}$$

Each policy's cost q_i equals the probability that the subsidy is paid: a dollar inside bankruptcy costs $q_i = F(\hat{v})$ in expectation, while a dollar outside bankruptcy costs $q_i = 1 - F(\hat{v})$. We leave the price of subsidies to unsecured debt unspecified, since they are paid only off the equilibrium path and our results do not depend on it.

5.2 Analysis of General Subsidies

5.2.1 Planner's Objective Is to Minimize Bankruptcy Costs

We first assume the firm has debt in place and the planner seeks to minimize the deadweight costs of bankruptcy. The planner's objective is

$$\mathscr{U}(\mathbf{s}) = -F(\hat{v}(\mathbf{s}))\,\mathbb{E}\big[(1-\lambda)v \mid v \leq \hat{v}(\mathbf{s})\big]. \tag{11}$$

The problem simplifies in two ways. First, because policies $\mathbf{s}$ enter only through the filing threshold, the planner effectively targets $\hat{v}$. This is equivalent to targeting the bankruptcy probability $F(\hat{v})$ and thus any associated externalities. Second, with a small budget $\varepsilon \to 0$, only marginal effects matter, so the planner minimizes

$$\left.\frac{d\hat{v}}{d\varepsilon}\right|_{\varepsilon=0} = \sum_i \frac{\partial \hat{v}}{\partial s_i}\left.\frac{ds_i}{d\varepsilon}\right|_{\varepsilon=0} = \sum_i -\left.\frac{\partial\Delta/\partial s_i}{\partial\Delta/\partial\hat{v}}\frac{1}{q_i}\right|_{\varepsilon=0}, \tag{12}$$

having used the chain rule to differentiate along each constraint.[39] The terms in the sum have a natural interpretation: Each is the product of "policy efficacy" × "bang for the buck." We now have the next result:

Proposition 4. *Table I (top rows) summarizes how subsidies affect the filing threshold $\hat{v}$, which the planner wants to minimize when $\lambda < 1$ and maximize when $\lambda > 1$.*

For $\lambda < 1$, subsidies to equity outside bankruptcy and to secured debt either inside or outside bankruptcy (or combinations of these) are welfare-equivalent and optimal; all other subsidies reduce welfare.

For $\lambda > 1$, subsidies to equity in bankruptcy are optimal as long as $F(\hat{v}) < 1/2$.[40]

The optimal policies are found by choosing the highest values from the 3rd row of Table I when $\lambda < 1$, and doing the opposite (choosing the lowest values) when $\lambda > 1$. When $\lambda < 1$, any subsidy that deters filing is equally effective: The planner can bribe equity not to file (s_E^O) or can bribe creditors to accept a distressed exchange (s_S^B or s_S^O).[41] But subsidies to equity in bankruptcy backfire by encouraging excessive filing.[42] Subsidies to unsecured debt likewise backfire by discouraging restructuring.

If bankruptcy is efficient ($\lambda > 1$), the same logic applies with signs reversed: When bankruptcy creates value, the optimal policies encourage filing and discourage restructuring. In particular, when the filing probability is low ($F(\hat{v}) < 1/2$), subsidies to equity in bankruptcy are optimal. When the filing probability is high, subsidies to unsecured creditors (in or out of bankruptcy) are optimal.

If the planner does not know whether λ is above or below one, it can implement the optimal

[39] The term $\partial\hat{v}/\partial s_i$ comes from the IC; $ds_i/d\varepsilon$ follows from the budget constraint.

[40] This is the typical case: bankruptcy probability is usually below one-half. These subsidies are also more attractive than policies that help only off the equilibrium path when $\lambda > 1$.

[41] The optimal policies induce the same filing threshold and differ only in how value is allocated across claimants. Subsidies to equity can benefit debt by reducing filing probability; subsidies to debt can benefit equity by facilitating deleveraging.

[42] Such subsidies also increase the value of priority and thus facilitate restructuring, but they do so at the cost of inducing bankruptcy—unlike subsidies to secured debt, which reduce filing and facilitate restructuring simultaneously.

policy by offering creditors a menu and allowing them to decide collectively (e.g., by vote) on what subsidy they would choose:

Proposition 5. *Define*

$$\pi(\lambda) := F(\hat{v}) \left(1 + \frac{1 - F(\hat{v})}{\hat{v} f(\hat{v})} \frac{D^*}{D_0}\right)\Bigg|_{\varepsilon=0}, \tag{13}$$

and suppose $\sup_{\lambda \in (0,1)} \pi(\lambda) \leq \pi(1) \leq \inf_{\lambda \in (0,1/(1-\theta))} \pi(\lambda)$.[43]

A welfare-maximizing planner can implement the optimal policy for any λ if secured creditors choose from the menu:

- *a subsidy to themselves inside bankruptcy at upfront price $1/\pi(1)$;*
- *a subsidy to equity inside bankruptcy.*

The menu induces the creditor collective to select the correct policy. Creditors are willing to pay more to avoid bankruptcy when it is costly. Thus, by Proposition 4, they pay for s_S^B when $\lambda < 1$ and choose s_E^B when $\lambda > 1$.

The key idea here is that creditors must pay a price in order to get a subsidy that facilitates restructuring. As the price increases, the subsidy to restructuring has less value. The price $1/\pi(1)$ is such that if bankruptcy and restructuring are equally efficient ($\lambda = 1$), then creditors are indifferent between (i) paying for the subsidy to themselves or (ii) selecting a subsidy to equity. They will therefore pay for the subsidy to themselves when bankruptcy is less efficient ($\lambda < 1$) but prefer a subsidy to equity when it is not ($\lambda > 1$), as the social planner would dictate.[44]

5.2.2 Planner's Objective Is to Maximize Debt Capacity

We now consider the planner's objective of maximizing the flow of credit by increasing the firm's ex ante debt capacity—how much the firm can borrow, anticipating restructuring (which occurs

[43] This condition depends on the equilibrium and is difficult to verify in general, but holds in examples such as the exponential distribution.

[44] This mechanism does assume that the planner can observe $\pi(1)$ when λ is unobservable.

for sure) and possible bankruptcy. (See Appendix D.3, especially equation (32).) In this case, the planner's objective is[45]

$$\mathscr{U}(\mathbf{s}) = \max_{D_0} \Big\{ F(\hat{v})\, s_U^B + (1 - F(\hat{v}))(D_0 + s_U^O) \Big\}. \tag{14}$$

Using the IC to eliminate D_0 yields

$$\mathscr{U}(\mathbf{s}) = \max_{\hat{v}} \Big\{ F(\hat{v})\mathbb{E}[\lambda\theta v + s_S^B \mid v \le \hat{v}] + (1 - F(\hat{v}))\big((1 - \lambda(1-\theta))\hat{v} + s_E^O - s_E^B + s_S^O\big) \Big\}. \tag{15}$$

Lemma 4. *Debt capacity does not depend on subsidies to unsecured debt (s_U^O and s_U^B).*

This result says that, as the firm chooses its initial debt optimally in anticipation of restructuring, it undoes any subsidies that accrue only in restructuring. Differentiating (15) and taking $\varepsilon \to 0$ gives the effects of other policies:

Proposition 6. *Table I (bottom rows) summarizes how subsidies affect debt capacity for $\lambda < 1$. Subsidies to equity outside bankruptcy and to secured debt inside or outside bankruptcy (or combinations thereof) are all optimal and have the same effect on debt capacity. Subsidies to equity in bankruptcy reduce debt capacity. For $\lambda > 1$, when bankruptcy creates surplus, the firm takes on so much debt that it always files, and subsidies to secured debt, which gets paid in bankruptcy, are optimal.*

Thus, for $\lambda < 1$, the same policies that maximize welfare ex interim (Proposition 4) also maximize debt capacity: Policies that create the most surplus ex post make creditors more willing to lend ex ante. There is a twist for $\lambda > 1$, when the planner wants to incentivize bankruptcy, because the firm can do that by increasing debt without changing the policy.

[45] The IC becomes:

$$F(\hat{v})s_U^B + (1 - F(\hat{v}))(D_0 + s_U^O) = F(\hat{v})\mathbb{E}[\lambda\theta v + s_S^B \mid v \le \hat{v}] + (1 - F(\hat{v}))\big((1 - \lambda(1-\theta))\hat{v} + s_E^O - s_E^B + s_S^O\big),$$

having substituted for D rather than $\hat{v}$ for convenience.

Corollary 1. *The planner can maximize debt capacity (for any λ) by subsidizing secured creditors inside bankruptcy.*

5.3 Specific Policies

Some instances of these general subsidies correspond to real-world policies and recent proposals. Subsidies to equity in bankruptcy (s_E^B) include policies that permit shareholders to retain ownership interests during a bankruptcy reorganization. For example, Congress recently amended small-business bankruptcy laws to permit reorganization plans that allow owners to retain their interests, as discussed in Morrison and Saavedra (2020). Proposition 4 suggests that, in our model, such policies are inferior to policies that subsidize secured creditors in bankruptcy, e.g., by extending them new credit at below-market rates, as in DeMarzo, Krishnamurthy, and Rauh's (2020) proposed DIPFF.

Many other policies are combinations of the general subsidies considered above. We can apply Proposition 4 to them as well, some of which we summarize below; any interpretations are for the $\lambda < 1$ case.[46]

1. **Value created in bankruptcy.** A long-standing question in bankruptcy reform is how to allocate value created in bankruptcy across secured debt, unsecured debt, and equity (cf. Adler (2015) and Baird (2015) with ABI (2014), Mann (1995), Janger (2015), and Jacoby and Janger (2014)). Our results suggest that this value should go entirely to secured creditors in bankruptcy. Other allocations backfire: directing value to unsecured debt discourages restructuring, and directing it to equity encourages excessive filing.

2. **Blanket asset subsidies.** A policymaker can inject cash to increase the firm's asset value. This type of policy has been applied to the the U.S. airline industry. After 9/11, for example, the Air Transportation Safety and Stabilization Act delivered \$5b to the airlines

[46] These and other policies are described in Casey and Posner (2015), Megginson and Fotak (2021), Kelly (2022), Goolsbee and Kreuger (2015), Levitin (2011), Buchholtz and Lawson (2021).

to compensate for operating losses. Similarly, during the pandemic, the Payroll Support Program made payments to the airlines to cover payroll. The incidence of the subsidy depends on whether the firm files for bankruptcy. If it does, the subsidy is split: A fraction $1-\theta$ goes to equity and a fraction θ to secured debt. If it does not, all debt is repaid, and the subsidy goes to equity.

3. **Stock purchases.** During the financial crisis, the federal government purchased preferred stock in AIG and General Motors. This policy is analogous to a blanket asset subsidy because it injects cash into the firm without increasing debt.

4. **Loan subsidies.** This policy has been used repeatedly to rescue failing firms. Loan guarantees were offered to Lockheed in 1971, Chrysler in 1980, and the airline industry following 9/11. Subsidized loans were offered to Chrysler and General Motors during the financial crisis. Direct loans were made to corporations during the pandemic. These policies are a combination of (i) asset subsidies because they increase cash and (ii) subsidies to unsecured creditors. The subsidy to unsecured debt is explicit in the case of loan guarantees (insuring against losses from default). Loan subsidies and direct loans are functionally equivalent to subsidies to unsecured debt because they increase D_0, making it harder to satisfy the IC in inequality (4).

5. **Debt purchases (and forgiveness).** A policymaker can purchase debt in the market and write it off, effectively paying a fair price to reduce the firm's debt. This resembles quantitative easing programs in which central banks purchase corporate debt, with the twist that the central bank does not enforce repayment on the purchased debt. In our model, such a decrease in unsecured debt, which is paid nothing in bankruptcy, is equivalent to a subsidy to secured debt outside bankruptcy, because it makes repayment more likely. $F(\hat{v})\mathbb{E}\left[\lambda\theta v | v < \hat{v}\right] \geq \big(1 - F(\hat{v})\big(D_0 - s_S^O - D\big)$, which is tantamount to a reduction in D_0.

6. **Restructuring subsidies.** A policy maker can "bribe" creditors to accept a restructuring, paying them conditional on participating in the exchange offer. One way to do this is to

alter the tax consequences of restructurings, as discussed in Campello, Ladika, and Matta (2018). Another is for the government to announce that it will effectively subsidize lenders who write down their loans, as discussed in Blanchard, Philippon, and Pisani-Ferry (2020). In our model, this corresponds to a subsidy to secured debt both in and out of bankruptcy. Thus, a dollar subsidy corresponds to $s_S^B = 1$ and $s_S^O = 1$ (and $s_i = 0$ otherwise).

The foregoing analysis reveals that commonly used policies are generally suboptimal because they include subsidies to equity in bankruptcy, which distort the filing decision (blanket asset subsidies and stock purchases) or because they effectively subsidize unsecured debt, which distorts the incentive to agree to a restructuring (loan subsidies). The only optimal policies are those that directly incentivize restructuring (restructuring subsidies) or reduce debt (debt purchases with forgiveness).

6 Extensions

We now relax several baseline assumptions. In the sections that follow, we allow (i) violations of priority in bankruptcy (e.g., value being distributed to junior creditors at the expense of seniors, or vice versa) and (ii) creditors to be concentrated rather than fully dispersed. Additional extensions appear in Appendix D; the Introduction provides a summary.

6.1 Priority Violations: LMEs, Secured Creditor Power, and Tort Claims

So far we have assumed that there are only two classes of debt with strict priority between them. This is a good approximation of reality, but not a perfect description of it. Sometimes junior claims are paid something before supposedly senior ones are paid in full, resulting in so-called "APR deviations." Sometimes, especially recently, a new class of "super-senior" debt is created, possibly diluting even secured debt in so-called "uptiering transactions." Similarly, some restructurings involve a "dropdown," in which previously-pledged collateral is moved to an "unrestricted subsidiary" (i.e. not subject to loan covenants), which then issues a new class of debt that effectively functions as super-senior debt. Priority violations can occur in other ways too. Senior creditors could exert control over the bankruptcy process and destroy firm value or redistribute value to themselves from junior creditors. Additionally, scholars and policymakers have advocated seniority for the claims of tort victims (whose claims are generally unsecured) over other creditors.

In this section, we develop an extension that captures these phenomena.

6.1.1 LMEs: Uptiers and Dropdowns

Liability Management Exercises (LMEs) have been increasingly common since they were used to restructure the first-lien debt of J.Crew in 2017 (via a "dropdown") and Serta Simmons in 2020 (via an "uptier"). For simplicity, we focus on uptiers, though the economic analysis is the same for dropdowns.

In the typical uptier, the firm secures consent from a majority of senior lenders to amend their loan agreement to permit issuance of new "super-senior" debt with priority over the existing senior debt. The majority then exchanges their existing senior debt for the new super-senior debt. The majority may take a write-down too (in Serta's case, for example, the majority took at 26% write-down). The result is that the majority lenders achieve priority over the remaining (minority) senior lenders (who continue to hold the original senior debt), albeit with a haircut. In addition, and simultaneously, the majority typically injects liquidity into the firm by making new loans with priority above all debt, including the super-senior debt. In Serta's case, for example, the majority lenders provided $200 million in new liquidity, which had the same priority as the super-senior debt. Uptiers, however, come with legal risk.[47] Many have been challenged in

[47] See, for example, the recent restructuring transactions and related litigation involving Serta Simmons, Boardriders, Envision Healthcare, Trimark, Wesco/Incora, and Hunkemoller, among others, as well as practitioner discussions of "creditor-on-creditor violence" and liability-management exercises. A recent academic analysis of uptiering is Buccola and Nini (2024).

lawsuits filed by minority lenders. If the minority lenders prevail, the potential remedies include reversing the uptier itself (i.e. returning all lenders to their priorities prior to the uptier) or requiring the firm or majority lenders to pay damages to the minority.

To study uptiering, we modify our baseline model in two respects.[48] The first is that there are now two potential rounds of restructuring, from D_0 to D_1 and then from D_1 to D_2. In each round, as in the baseline, the new debt is supposedly senior to the old debt. The second is that, consistent with practice, the second round is only offered to a subgroup of creditors of mass ϕ (the "majority") and the seniority of the new debt is disputed. To capture the asymmetric offer, let D_2^{ϕ} be the face value of senior debt offered to the majority, so the total amount of debt becomes $D_2 := \phi D_2^{\phi} + (1-\phi)D_1$. To capture the legal uncertainty,[49] we assume that new senior debt is paid first with probability ρ and reverts to the old debt D_1 (i.e. the new debt is voided, and the majority returns to sharing pro rata with the minority) with probability $1-\rho$.[50] Equity still receives $(1-\theta)\lambda v$ in bankruptcy; what changes is how $\theta\lambda v$ is split between new and old debt.[51] For simplicity, we assume the mass of the majority is not too small, so the minorities recover nothing in bankruptcy $\phi D_2^{\phi} \geq \lambda\theta\hat{v}(D_2)$, or equivalently $\phi \geq \underline{\phi} := 1 - \frac{D_2}{D_1}\frac{1-\lambda}{1-\lambda(1-\theta)}$.

We solve backward through the two rounds of restructuring. In the second round, the creditors

[48] Antill, Jiang, and Wang (2025) also study LMEs but find, in contrast, that they can be inefficient, due to frictions not in our model, notably asymmetric information and illiquidity, suggesting their results apply better to smaller firms, where those frictions are likely to be first order. Symmetrically, ours probably apply more to larger firms, where we think our focus on dispersed creditors and insolvency is appropriate, as discussed in in footnote 24.

[49] The parameter ρ captures legal and institutional uncertainty regarding the enforceability of uptiering transactions, including uncertainty about whether courts will uphold the contractual seniority of super-senior claims created outside of bankruptcy.

[50] Another possible resolution of the dispute is to recognize the superpriority but require the majority to pay compensatory damages to the minority. We show later that the two remedies achieve the same outcomes.

[51] Although we interpret creditor power as a policy parameter (capturing features of the Bankruptcy Code or judicial tendencies), it may also reflect market structure. Jiang, Li, and Wang (2012) find that hedge-fund ownership of unsecured debt is associated with higher total creditor recoveries (a higher θ) but also higher unsecured recoveries (a lower ρ). Our analysis therefore highlights a potential downside of hedge-fund participation: It can impede restructuring (Section 4.3).

accept a write-down from D_1 to D_2^ϕ if

$$\begin{aligned}
&\big(1-F(\hat{v}(D_2))\big)D_2^\phi + F(\hat{v}(D_2))\mathbb{E}\left[\rho\frac{\lambda\theta v}{\phi} + (1-\rho)\lambda\theta v \mid v < \hat{v}(D_2)\right] \\
&\quad \geq \big(1-F(\hat{v}(D_2))\big)D_1 + F(\hat{v}(D_2))\mathbb{E}[\rho \times 0 + (1-\rho)\lambda\theta v \mid v < \hat{v}(D_2)]\,.
\end{aligned} \tag{16}$$

Relative to the baseline IC in equation (4), a hold-out creditor (the "minority") now receives a positive bankruptcy payoff with probability $1-\rho$. Substituting $D_2^\phi = D_1 - \frac{D_1-D_2}{\phi}$ into the IC above and simplifying yields

$$D_1 - D_2 \quad \leq \frac{F(\hat{v}(D_2))}{\big(1-F(\hat{v}(D_2))\big)}\mathbb{E}[\rho\lambda\theta v \mid v < \hat{v}(D_2)]\,. \tag{17}$$

Notice that ϕ has vanished from the IC, giving the next result:

Lemma 5. *Given the outcome of the first round of restructuring, D_1,*

- *the aggregate debt level D_2 and all payoffs—to each group of creditors and the firm—are independent of the size of the majority ϕ;*
- *the expected payoffs to majority and minority creditors coincide.*

Minority creditors, who have been active in court, might be aggrieved at our finding that they are not worse off than the majority.[52] Our finding also contrasts with much of the existing academic and practitioner commentary, which views uptiers and other LMEs as a form of

[52]Our result is not necessarily in tension with the observation that market prices of secured debt (held by the minority) often fall after an uptier: It merely says the minorities are not worse off *compared to the majority*, who takes a write-down.

To be sure, the minority could suffer harm for reasons outside our model. One is that LMEs typically include not only debt restructuring but other corporate actions, such as issuing new senior debt, that dilutes the minority—an uptier is often part of a broader transaction that includes new liquidity with priority over all debt, including the uptiered super-senior debt. Another is that the majority could collude with the firm to expropriate value from the minority. Yet another is that the bankruptcy costs $(1-\lambda)$ could increase as a result of the litigation induced by the LME itself (an idea that suggests it is the legal uncertainty about how to resolve the transaction that is the problem, not the transaction itself).

Thus, our model suggests that while the minority might have good reason to seek damages after an uptier, it would not be due to the uptier itself, but rather due to omitted variables, market power, or legal costs, for example.

"creditor-on-creditor violence." The intuition for our result is that the firm will make the lowest possible offer that a marginal creditor will accept: The creditor will be indifferent between accepting (getting the majority payoff) and holding out (getting the minority payoff). It breaks even only because it is better off when the firm enters bankruptcy. By contrast, when the firm survives the distress, the minority is better off by not having taken a write-down. The rent of being in the majority erodes as groups of creditors compete to be the "majority," as they did in Serta.[53]

Another important element of our result is that the size of the majority is irrelevant. This is because creditors are indifferent between different ways of achieving the same total debt level. To see this, note that for any given aggregate write-down, a creditor is indifferent between taking a dollar write-down for sure versus taking an n-dollar write-down $1/nth$ of the time. Or, by the law of large numbers, a creditor is indifferent (in expectation) between a dollar write-down with certainty versus being part of $1/nth$ of creditors (the majority) taking an n-dollar write-down. Put differently, for each majority creditor, for every dollar of write-down $(\frac{D_1-D_2}{\phi})$, he gets proportionally more in bankruptcy $(\rho\frac{\lambda\theta v}{\phi})$ and the effect of size ϕ washes out. Similarly, for a minority creditor, each dollar of write-down by the majority increases the minority creditor's probability of full repayment (no haircut), which offsets the minority's reduced recovery in bankruptcy, and again the effect of size washes out.

Because of this point in Lemma 5, we can set $\phi = 1$ below without affecting the analysis. Additionally, because condition (17) mirrors equation (5) except for the bracketed term, which is increasing in ρ, we obtain the next result:

[53] In a brief filed in related litigation, the process leading to the Serta uptier is described as follows: "To achieve the best possible financing terms, the Committee [appointed by Serta's Board of Directors] established a competitive process. Serta's agent solicited proposals from eleven different lending groups, seven of which expressed interest. Simultaneously, several lending groups ... reached out to Serta unsolicited. All told, Serta discussed potential transactions with lenders holding 70% of Serta's outstanding first-lien debt under the Term Loan Agreement, as well as various lenders who were not parties to that Agreement. The Participating Lenders initially proposed a pro-rata transaction that would be available to all first-lien lenders, but the Competing Lenders proposed a non-pro-rata transaction. The Participating Lenders ... made a competing offer for a non-pro-rata transaction." (Plaintiff Lenders and Represented Third-Party Defendants' Motion to Dismiss LCM Defendants' Amended Counterclaims and Third-Party Claims (*In re Serta Simmons Bedding, LLC*), Adversary Proc. No. 23-09001 (ECF 387) (2025).)

Lemma 6. *Strict enforcement of seniority ($\rho = 1$) maximizes the feasible write-down in* (17) *and therefore facilitates restructuring in the second round.*

We now proceed backwards to the first round of restructuring. We assume that the firm can exclude first-round holdouts from the second round. That is, the second round of restructuring will apply only to lenders—holding a fraction ϕ of the original debt—that restructure and obtain secured debt in the first round. In other words (and consistent with actual practice) any unsecured debt that dissents from a first-round restructuring will remain unsecured during the second round. Creditors' first-round IC is therefore:

$$(1 - F(\hat{v}(D_2)))D_1 + (1 - \rho)F(\hat{v}(D_2))\mathbb{E}[\lambda\theta v | v < \hat{v}(D_2)] \geq (1 - F(\hat{v}(D_2))D_0. \tag{18}$$

Applying Lemma 5, we can set $\phi = 1$, which implies that $D_2 = D_2^{\phi=1}$. Now, if we add the ICs from the first and second rounds, we have a striking result:

$$\big(1 - F(\hat{v}(D_2))\big)D_2 + F(\hat{v}(D_2))\mathbb{E}[\lambda\theta v \mid v < \hat{v}(D_2)] \geq (1 - F(\hat{v}(D_2))D_0. \tag{19}$$

This is just the IC of the baseline model (equation (4)). We therefore have the following result:

Proposition 7 (Irrelevance of Uptiering)**.** *The equilibrium debt level, D_2, after two rounds of restructuring, coincides with the level that would be observed if there were only one round. In particular, the equilibrium debt level depends neither on ρ (the probability that new debt is senior) nor on ϕ (the fraction of creditors included in the majority). In addition, the expected payoffs of the firm and the creditors, majority or minority, also coincide with the payoffs if there were only one round.*

This is a type of "Modigliani–Miller" irrelevance proposition for uptier transactions: Whether the firm has a second round of restructuring, to whom the second round is offered, and whether the super-priority of the second offer is recognized, are irrelevant. The source of the first irrelevance is the fact that creditors anticipate multiple rounds of restructuring. In the first round, they know their debt will be restructured again in the second. They value their claims (correctly) as if the final debt level will be D_2 and, as a result, additional rounds do not facilitate further write-downs. The source of the second irrelevance stems from priority being more valuable when only offered to a subgroup, and this subgroup is willing to take steeper write-downs, which exactly offset the higher outstanding debt of the minority. The third irrelevance holds because creditors correctly anticipate the probability of the super-priority being recognized, and adjust the value of the priority offered in the first round.

We note, however, that the dynamics of restructuring do depend on the probability of effective subordination ρ. Indeed, if priority is perfectly respected ($\rho = 1$), there is no restructuring in the first round (you know you will be completely diluted in the second), and if it is completely disregarded ($\rho = 0$), there is no scope for restructuring in the second:

Corollary 2. *The write-down in the first round $D_0 - D_1$ is decreasing in ρ.*

- *If $\rho = 1$, there is no write-down in the first round: $D_1 = D_0$.*
- *If $\rho = 0$, there is no write-down in the second round: $D_2 = D_1$.*

This result shows how the creditors adapt their response in the first round to the possibility of enforceable uptiers in the second: As the probability that courts enforce an uptier increases, creditors attach a lower value to priority offered in the first round, and are less willing to accept a write-down.[54]

So far, we have assumed that, if a court invalidates an uptier, the super-senior debt is eliminated and the original debt reinstated. Alternatively, a court could enforce the super-senior debt, but require the majority to pay compensatory damages $\tau(v)$ to each minority creditor. This would result in a total transfer of $(1-\phi)\tau(v)$ to the minority, borne equally by the majority of mass ϕ. We do not require $\tau(v)$ to be the exact damage inflicted on the minorities: Any

[54]This result also explains why the market prices of leveraged loans declined after market actors became more aware of the possibility of an LME, as Ivashina and Vallée (2018) show. A sudden increase in ρ means that senior lenders expect a larger write-down in a second restructuring, which reduces the value of existing senior debt.

transfer works.[55]

Proposition 8 (Equivalence of Legal Remedies)**.** *The equilibrium aggregate debt level D_2 and the welfare of all creditors and the firm are identical for any transfer $\tau(\cdot)$. And they are the same as if there were never a second round, or if the uptier transaction were reverted. The debt dynamic is also the same: A higher ρ increases the second-round write-down but decreases the first-round write-down.*

This result suggests that the remedy applied by the court is irrelevant, provided it can be correctly anticipated, reminiscent of Coase's theory that the initial allocation of property rights is irrelevant as long as it is clearly defined. Competition for priority will erode the benefits of priority, leaving the minority no worse off.

6.1.2 Secured Creditor Control

The discussion so far abstracts from at least one contestable issue with respect to secured creditor power—namely, that secured creditors could exercise control in ways that destroy firm value, such as by forcing fast sales at fire-sale prices to ensure repayment for themselves (Ayotte and Ellias (2020), Antill (2020), Ayotte and Morrison (2009)). To capture this, Internet Appendix H allows secured creditors to influence bankruptcy outcomes. We show that secured creditor control can either facilitate or hinder restructuring depending on its effect on total payoffs. If secured creditors divert value from juniors without hurting equity (as in Ayotte and Ellias (2020)), restructuring becomes easier. But if they induce inefficient liquidation that lowers payoffs for all claimants (Ayotte and Morrison (2009); Antill (2020)), restructuring becomes harder.

6.1.3 Tort Claimants

Debates over priority also concern tort claimants ("accidental" or "involuntary" creditors). Appendix D.4 shows that, to facilitate restructuring, tort claims should rank above junior debt but

[55] In fact, the transfer $\tau(v) = \lambda\theta v$ implements the special case in which the uptiers were invalidated.

below senior debt. This increases the payoff gap between senior and junior debt, raising the value of priority and improving the feasibility of restructuring.

6.2 Concentrated Debt Holdings and Debt-Equity Exchanges

Thus far we have assumed dispersed creditors, which implies that feasible restructurings involve swapping junior debt for senior debt. This fits most corporate exchange offers but cannot explain observed cases involving debt-for-equity swaps (e.g., Asquith, Gertner, and Scharfstein (1994)). We now allow for creditor concentration and show that debt-equity exchanges arise if and only if debt holdings are sufficiently concentrated.

Here we suppose that the firm's debt is held by a single large creditor (or coordinating group) with probability ξ; otherwise, the debt is dispersed per the baseline. We assume that the firm makes its offer without knowing whether it faces a large or dispersed creditor, but creditors know the true structure when deciding whether to accept.[56]

When creditors are dispersed, the baseline IC (4) applies and debt–equity swaps are infeasible. When the creditor is concentrated, however, the IC becomes:

$$\begin{aligned} &F(\hat{v}(D))\,\mathbb{E}\big[\big(\theta + (1-\theta)(1-\varphi)\big)\lambda v \mid v \leq \hat{v}(D)\big] \\ &\quad + \big(1 - F(\hat{v}(D))\big)\mathbb{E}[D + (1-\varphi)(v - D) \mid v \geq \hat{v}(D)] \\ &\qquad \geq F(\hat{v}(D_0))\,\mathbb{E}[\lambda\theta v \mid v \leq \hat{v}(D_0)] + (1 - F(\hat{v}(D_0)))D_0. \end{aligned} \tag{20}$$

The LHS is the payoff from accepting a mix of senior debt D and equity $(1-\varphi)$; the RHS is the payoff from rejecting. Unlike dispersed creditors, a concentrated creditor internalizes the fact that rejecting causes the restructuring to fail.

The firm, anticipating either creditor structure, can pursue three strategies:

1. offer just enough senior debt to make dispersed creditors accept (no equity);

[56] Debt concentration often changes as firms enter distress; ξ can be interpreted as the firm's belief about future concentration when the offer is made.

2. offer just enough equity to make the large creditor accept (no debt);

3. offer enough of both senior debt and equity to make *both* types accept (leaving some rent to dispersed creditors).

The firm faces a trade-off: A universally acceptable offer (option 3) minimizes the chance of bankruptcy but is most generous to creditors; more selective offers (options 1–2) reduce rents but increase the probability of distress. Comparing payoffs yields the following result:

Proposition 9. *Suppose D_0 is sufficiently large. Then there exist thresholds $\tilde{D}$, $\underline{\xi}$, and $\bar{\xi}$ (given in* (93), (115), (116)*) such that:*

- *If $D^* \geq \tilde{D}$ and $\underline{\xi} < \bar{\xi}$, there are three regions:*
  - *If $\xi \leq \underline{\xi}$, the firm offers senior debt only (accepted only by dispersed creditors).*
  - *If $\underline{\xi} < \xi \leq \bar{\xi}$, the firm offers both senior debt and equity (accepted by all creditors).*
  - *If $\xi > \bar{\xi}$, the firm offers equity only (accepted only by the concentrated creditor).*
- *Otherwise, there are two regions: below a threshold, the firm offers senior debt (accepted by all), and above it the firm offers equity only.*

In sum, the baseline model is robust to modest concentration, but sufficiently concentrated holdings induce the firm to use equity. This helps explain why debt-equity swaps arise in practice. The result aligns with James (1995), who finds that banks (unlike dispersed bondholders) often take equity in restructurings because they internalize the full effect of write-downs; see also Jostarndt and Sautner (2009).

7 Conclusion

We develop a model based on two observations about restructuring with dispersed creditors. First, restructuring is difficult—often impossible—unless the firm can dilute existing debt with high-priority (senior) debt in an exchange offer. Second, priority is valuable only to the extent that bankruptcy is likely: Creditors do not care about seniority if bankruptcy will never occur, but they care a great deal when bankruptcy is likely, and the value of priority rises with the probability of filing. Although intuitive, this yields a counterintuitive implication: Bankruptcy and restructuring are complements. Policies that make bankruptcy attractive to equity holders also facilitate restructuring. This complementarity drives our main results on how deadweight costs and creditor friendliness affect restructuring.

Our model helps guide policymakers seeking to assist distressed firms via bankruptcy reforms or direct interventions. For example, our calculations for the U.S. suggest that reducing creditor friendliness θ would likely increase restructuring. The model also suggests that policymakers should favor policies that facilitate restructurings (e.g., rewarding creditors for accepting write-downs) and those that increase senior lenders' payoff in bankruptcy (e.g., government backstops to DIP loans). Strikingly, such policies are optimal regardless of whether the objective is to minimize bankruptcy costs ex interim or to maximize debt capacity ex ante.

Our analysis also speaks to the design of bankruptcy policy outside of crises. For instance, the APR can either aggravate or mitigate distress. It facilitates restructuring when it helps secured creditors maintain priority over other debt, but it can hinder restructuring if it prevents equity from receiving value in bankruptcy.

The framework sheds light on ongoing controversies. One concerns liability management exercises (LMEs), which are often viewed as a form of "violence" in which majority secured creditors benefit at the expense of minority secured creditors. We show, in the context of uptiers, that LMEs cause no harm (or benefit), as long as they are anticipated. Indeed, the possiblity of an uptier is irrelevant: The expected payoffs of the firm and its creditors (majority or minority) are the same, regardless of whether an uptier is possible or not.

Our framework also sheds light on controversies arising from "Debtor in Posession" (DIP) loans in bankruptcy, which are typically provided by existing senior creditors and criticized for being highly profitable (e.g., Eckbo et al. (2019)) or giving senior lenders excessive control in bankruptcy (e.g., Ayotte and Morrison (2009)). Our model offers another perspective: DIP loans increase senior lenders' payoff in bankruptcy and protect them from dilution, thereby making restructuring more feasible and avoiding distress costs.

Finally, the model provides insight into the evolution of U.S. bankruptcy practice. In the late 19th century, before modern bankruptcy statutes, lawyers developed reorganization devices such as the "equity receivership," a proto–Chapter 11 procedure that often (i) maximized secured creditor recoveries, (ii) preserved value for equity, and (iii) eliminated payoffs to unsecured creditors (Miller and Berkovich (2006)). In retrospect, this private ordering resembled what our model deems efficient in facilitating restructuring.

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Type of subsidy: s_i		Inside Bankruptcy			Outside Bankruptcy		
		Equity s_E^B	Secured s_S^B	Unsecured s_U^B	Equity s_E^O	Secured s_S^O	Unsecured s_U^O
[1] Cost/Probability of subsidy:	q_i	$F(\hat{v})$	$F(\hat{v})$	$F(\hat{v})$	$1-F(\hat{v})$	$1-F(\hat{v})$	$1-F(\hat{v})$
[2] Effect of subsidy on welfare:	$\left.\frac{\partial\Delta}{\partial s_i}\right\vert_{\varepsilon=0}$	$-(1-F(\hat{v}))$	$F(\hat{v})$	$-F(\hat{v})$	$1-F(\hat{v})$	$1-F(\hat{v})$	$-(1-F(\hat{v}))$
[3] Adjusted for probability of subsidy:	$-\left.\frac{\mathrm{d}\hat{v}}{\mathrm{d}\varepsilon}\frac{\partial\Delta}{\partial\hat{v}}\right\vert_{\varepsilon=0} = \frac{1}{q_i}\left.\frac{\partial\Delta}{\partial s_i}\right\vert_{\varepsilon=0}$	$-\frac{1-F(\hat{v})}{F(\hat{v})}$	1	-1	1	1	-1
[4] Effect of subsidy on debt capacity:	$\left.\frac{\partial L(\mathbf{s})}{\partial s_i}\right\vert_{\varepsilon=0}$	$-(1-F(\hat{v}))$	$F(\hat{v})$	0	$1-F(\hat{v})$	$1-F(\hat{v})$	0
[5] Adjusted for probability of subsidy:	$\left.\frac{\partial L(\mathbf{s})}{\partial\varepsilon}\right\vert_{\varepsilon=0} = \frac{1}{q_i}\left.\frac{\partial L(\mathbf{s})}{\partial s_i}\right\vert_{\varepsilon=0}$	$-\frac{1-F(\hat{v})}{F(\hat{v})}$	1	0	1	1	0

Table I: Welfare effect and debt capacity of general subsidies. (The 2nd and 4th rows represent the effects on the welfare and the debt capacity of one marginal dollar, and the 3rd and 5th rows adjust them for the probability of the subsidies, calculated by dividing the 2nd and 4th rows by the 1st row, the cost of a one-dollar subsidy. Higher values of $-\left.\frac{\mathrm{d}\hat{v}}{\mathrm{d}\varepsilon}\frac{\partial\Delta}{\partial\hat{v}}\right\vert_{\varepsilon=0}$ correspond to higher welfare; see equation (12), but note that we multiply by the denominator $\partial\Delta/\partial\hat{v}$ for brevity, as it is the same for all policies. Similarly, higher values of $\frac{1}{q_i}\left.\frac{\partial L(\mathbf{s})}{\partial s_i}\right\vert_{\varepsilon=0}$ correspond to higher debt capacity.)

$\mathbf{s}$	Value to secured debt	Blanket assets	Assets in bankruptcy	Restructuring	Debt purchases
$-\frac{\mathrm{d}\hat{v}}{\mathrm{d}\varepsilon}\frac{\partial\Delta}{\partial\hat{v}}\Big\vert_{\varepsilon=0}$	1	θ	$1-\frac{1-\theta}{F(\hat{v})}$	1	1

Table II: The welfare effects of specific subsidies: The entries correspond to $\nabla\hat{v}\cdot\mathbf{s}$ for each policy $\mathbf{s}$ as described in Section 5.3.

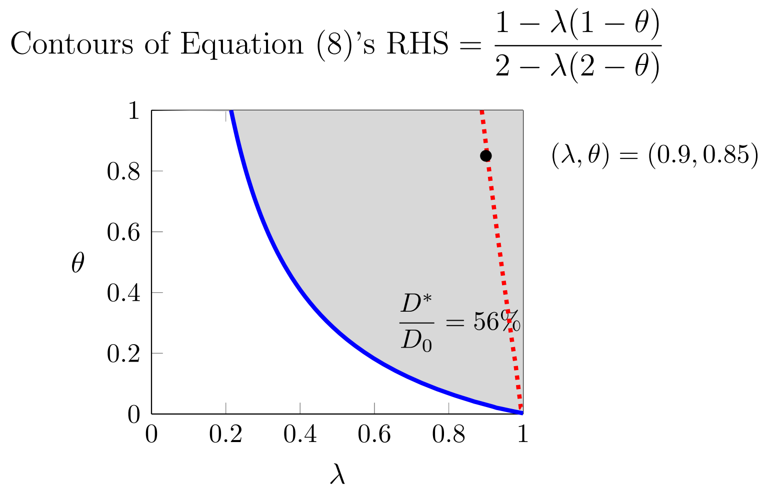

Figure 1: Contours of equation (8)**'s RHS:** The figure shows two contours of equation (8)'s RHS. The first (solid blue) corresponds to our empirical estimate of the LHS, $D^*/D_0 = 56\%$. The second (dotted red) corresponds to our empirical estimates of $(\lambda, \theta) = (0.9, 0.85)$, i.e. RHS $= \frac{1-0.9\times(1-0.85)}{2-0.9\times(1-0.85)} \approx 90\%$. The grey region is the set of (λ, θ) for which the condition is satisfied given our estimate of D^*/D_0. That the region contains our estimate of (λ, θ) indicates that the condition is satisfied empirically.

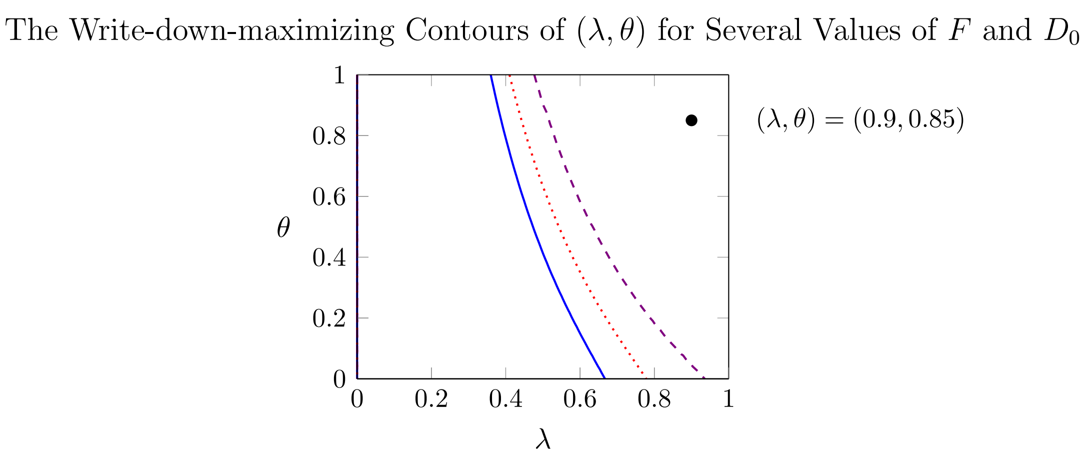

Figure 2: The contour sets of (λ, θ) **s.t.** $\mathbf{d}D^*/\mathbf{d}\theta = 0$**:** The solid blue curve corresponds to $F \sim \mathrm{U}[0, 2]$ and $D_0 = 1$; the dotted red to $F \sim \mathrm{U}[0, 3]$ and $D_0 = 1$; the dashed purple to $F \sim \mathrm{U}[0, 10]$ and $D_0 = 1$. Condition (8) is satisfied for (λ, θ) to the northeast of the curves. Our estimate $(\lambda, \theta) = (0.9, 0.85)$ is within that region in all cases.

Appendix

A Out-of-court Liquidation and Bargaining in Bankruptcy

In the baseline model, we assume that the firm either repays its debt or files for bankruptcy. In practice, it can also default without filing. In this case, creditors have the right to liquidate assets.[57] Here we model these two options explicitly.

1. **Liquidation.** If the firm defaults and does not file for bankruptcy, creditors can seize the firm's assets. We assume that their liquidation (or redeployment) value is less than the value to incumbent equity holders,[58] leading to deadweight costs $(1-\mu)v$. All of the remaining value μv goes to creditors; equity holders get nothing. Creditors also have the option to liquidate if bargaining with equity holders breaks down after a bankruptcy filing, as we describe next.

2. **Bankruptcy.** To avoid liquidation, the firm can file for bankruptcy. We assume the same bankruptcy costs as in the baseline model. The value of the firm net of bankruptcy costs, λv, is distributed through a structured bargaining process.[59] In particular, creditors can insist on a payoff no less than their recovery in a bankruptcy liquidation, $\mu\lambda v$ (this is called the "best interests test" and represents an outside option during the bargaining). The surplus created by avoiding liquidation, $\lambda v - \mu\lambda v$, is split between the parties. The

[57] One thing we abstract from here is that creditors can file an "involuntary" bankruptcy case against a firm. Under U.S. law, they must prove that the firm is "generally not paying such debtor's debts as such debts become due" 11 U.S.C. §303(h)(1). Courts have not given a precise or consistent definition of "generally not paying," but it appears to describe a situation where the firm has defaulted on multiple debts that account for a substantial fraction of total debt (Levin and Sommer (2020)). This is a situation close to insolvency, that is, $v \leq D$. Equation (3), however, shows that the firm will choose to file when $v \leq \hat{v}(D)$. As discussed above, $\hat{v}(D)$ will exceed D whenever θ and λ are greater than 0. This suggests that creditor power to start a case is relevant only in the (unusual) situation where the bankruptcy law offers no payout to equity or has no deadweight costs. In practice, involuntary filings account for about two percent of corporate bankruptcy filings (Hynes and Walt (2020)).

[58] For the microfoundations of this wedge in value, see, e.g., Aghion and Bolton (1992), Hart (1995), and Shleifer and Vishny (1992). For evidence on the deadweight costs of liquidation, relative to reorganization, see Bernstein, Colonnelli, and Iverson (2019).

[59] See Bisin and Rampini (2005) and von Thadden, Berglöf, and Roland (2010) for models rationalizing the institution of bankruptcy. See Waldock (2020) for a comprehensive empirical study of bankruptcy filings by large corporations in the U.S.

split is a function of various bankruptcy rules that allocate bargaining power to creditors in some cases (e.g., rules governing adequate protection and lift-stay motions) and equity holders in others (e.g., deviations from absolute priority, third-party releases, and, more broadly, management retention agreements). We model this bargaining environment using the generalized Nash bargaining protocol: Creditors get their liquidation value $\mu\lambda v$ plus a fraction $\hat{\theta}$ of the surplus created by avoiding liquidation, where $\hat{\theta}$ is their bargaining power.

When a firm reorganizes in bankruptcy, creditors bargain collectively and are guaranteed (via the "best interests test") a payoff no lower than what they would receive in a liquidation ($\mu\lambda v$). The extent to which their payoff exceeds $\mu\lambda v$ depends on the value available for distribution in a reorganization (λv) and their bargaining power ($\hat{\theta}$). Thus,

$$\begin{aligned}
\text{creditors' payoff} &= \text{liquidation value} + \hat{\theta} \times \text{surplus from reorganization} && (21)\\
&= \mu\lambda v + \hat{\theta}\big(\lambda v - \mu\lambda v\big) && (22)\\
&= \big(\mu + (1-\mu)\hat{\theta}\big)\lambda v && (23)\\
&\equiv \lambda\theta v, && (24)
\end{aligned}$$

where $\theta := \mu + (1-\mu)\hat{\theta}$ is the "creditor friendliness" of the baseline model (Section 2.2), now expressed as a combination of the value of creditors' outside option (μ) and their direct bargaining power in bankruptcy court ($\hat{\theta}$).

The creditors' payoff above corresponds to the repayment they receive after bankruptcy reorganization. So $D - \lambda\theta v$ can be interpreted as the write-down creditors take *in bankruptcy*. Thus the next lemma:

Lemma 7. *The in-bankruptcy write-down is decreasing in λ and θ.*

Contrasting this result with our main comparative statics (Proposition 2 and Proposition 3) is tantamount to contrasting the literature in which bankruptcy is an outside option in the bilateral renegotiation between the firm and a single (e.g., Corbae and D'Erasmo (2021)) with our model of bankruptcy as an "inside option" for dispersed creditors. The difference is that a large creditor's action determines the outcome—rejecting a restructuring causes bankruptcy—whereas a small creditor's does not—rejecting a restructuring has a negligible effect on the likelihood of bankruptcy, given others' actions. Per Lemma 7, switching from coordinated to uncoordinated creditors reverses the effect of bankruptcy costs $(1-\lambda)$ and introduces ambiguity in the effect of creditor friendliness (θ).

B Ex Post Restructuring

In our baseline model, we allow for restructuring only at Date 0, abstracting from it at Date 1. Here, we show this assumption is without loss of generality because restructuring is generally infeasible ex post, when there is no uncertainty about firm value:

Lemma 8. *There is no ex post restructuring that (uncoordinated) creditors are willing to accept and that the firm is willing to offer.*[60]

Proof. We consider equity and debt restructuring in turn.

Equity restructuring. There are two cases corresponding to $v \lesseqgtr \hat{v}(D)$. In each case, each individual creditor must be better off accepting fraction $1-\varphi$ of the equity than keeping their debt and equity holders must be better off offering it than not.

In either case, there is no bankruptcy following a successful restructuring, so an individual creditor, taking others' acceptance as given, must be better off accepting than getting repaid in full:

$$(1-\varphi)v \geq D. \tag{25}$$

Case 1: $v \geq \hat{v}(D)$. In this case, there is no bankruptcy even absent a restructuring, so equity

[60] To be precise, change in debt that does not affect anything else—it leaves payoffs and actions unchanged—could be possible. For simplicity, we ignore such immaterial debt changes.

must be better off offering it than repaying in full:

$$\varphi v > v - D. \tag{26}$$

This is mutually incompatible with creditors' accepting (condition (25)).

Case 2: $v < \hat{v}(D)$. In this case, the firm would file for bankruptcy absent a restructuring. So equity holders must prefer their residual claim on the firm φ after a restructuring to what they get in bankruptcy absent one, $(1-\theta)\lambda$:

$$\varphi v \geq (1-\theta)\lambda v. \tag{27}$$

This is mutually incompatible with creditors' accepting (condition (25)) and the assumption that $v < \hat{v}(D)$.

Debt restructuring. Denote the total amount of debt after a potential successful restructuring to $D' < D$. Again, there are two cases, now corresponding to $v \lesseqgtr \hat{v}(D')$.

Case 1: $v \geq \hat{v}(D')$. In this case, an individual creditor accepts the restructuring whenever its payoff D' from accepting exceeds its payoff D from holding out, or $D' \geq D$, i.e. it never accepts a write-down.

Case 2: $v < \hat{v}(D')$. In this case, the firm files for bankruptcy even if a restructuring is successful. Thus, it is payoff equivalent to not restructuring: The firm gets $(1-\theta)\lambda v$ and creditors get $\lambda\theta v$.[61] □

Ex post restructuring is subject to such a severe hold-out problem that no debt reduction is feasible. The reason is that an effective restructuring reduces debt enough that the firm repays its debt for sure—all uncertainty being resolved, the probability that the firm files for bankruptcy

[61]Note: If we allowed for mixed equilibria, one could exist in this case, albeit an unrealistic one. If the firm could commit to file randomly, say with probability p, then it could induce the creditors to accept. Such random filing is time consistent only if $\hat{v}(D') = v$, i.e. if the firm is indifferent between filing and not conditional on the offer D' being accepted. This strategy is implemented with $D' = \big(1-\lambda(1-\theta)\big)v$ (which makes the firm indifferent to filing if the offer is accepted) and $p = \frac{D_0 - D'}{D_0-(1-\lambda)v}$ (which makes the creditors indifferent to accepting it).

is zero. Thus, a hold-out creditor anticipates being repaid in full with certainty. And no creditor is willing to accept a write-down on its debt if its debt is valued at par. Hence, any ex post restructuring is doomed to fail.

C Example: Optimal Creditor Friendliness if v Is Uniform

Here, we illustrate Proposition 3 for v uniform, $F(v) \equiv v/\bar{v}$ on $[0, \bar{v}]$. In this case, creditors' binding IC ($\Delta = 0$ in equation (4)) becomes

$$\frac{\lambda\theta}{2\bar{v}}\big(\hat{v}(D^*)\big)^2 = (D_0 - D^*)\left(1 - \frac{\hat{v}(D^*)}{\bar{v}}\right). \tag{28}$$

Substituting for $\hat{v}$ from equation (3) and solving gives[62]

$$D^* = \frac{\big(1-(1-\theta)\lambda\big)\bar{v} + D_0 - \sqrt{\Big(D_0 - \big(1-(1-\theta)\lambda\big)\bar{v}\Big)^2 + 2\lambda\theta\bar{v}D_0}}{2 - \frac{\lambda\theta}{1-(1-\theta)\lambda}}. \tag{29}$$

This expression illustrates that the write-down $D_0 - D^*$ is increasing in λ, as per Proposition 2, and is hump-shaped in θ, as per Proposition 3; see Figure 3.

D Further Extensions

D.1 Court Congestion

We have assumed thus far that the costs of bankruptcy $(1-\lambda)v$ do not depend on whether restructuring occurs. This is reasonable for an individual firm because a single restructuring is unlikely to affect the efficiency of courts. However, taken in aggregate, restructurings can affect

[62] The other root of the quadratic is larger than D_0 and is omitted.

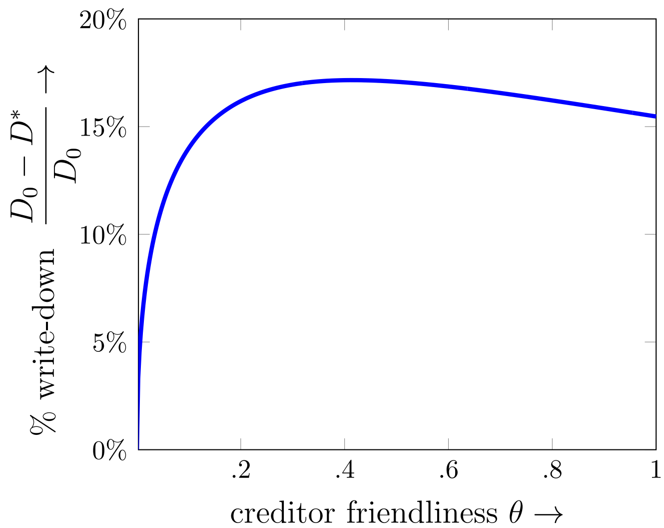

Figure 3: The percentage write-down for uniform v on $[0, \bar{v}]$, with $\bar{v} = 100$, $D_0 = 50$, and $\lambda = 1/2$.

the costs of bankruptcy—if there are more out-of-court restructurings, fewer firms will file for bankruptcy, and courts are likely to be less congested. In other words, the costs of bankruptcy could be increasing in the aggregate number of firms that file. Here we show that this effect can create a feedback loop, amplifying the effects of bankruptcy costs on the hold-out problem.

We assume that there is a unit of ex ante identical firms with independent asset values, and we assume that the costs of bankruptcy increase with the number of firms that file (which equals the probability that any individual firm files $F(\hat{v})$, by the law of large numbers). We assume that courts can process a maximum number of filings (κ) before experiencing "congestion costs" (see, e.g., Iverson (2018)). Specifically, we assume that

$$\text{bankruptcy costs } = 1 - \lambda_H - \mathbb{1}_{\{F(\hat{v})>\kappa\}}(\lambda_L - \lambda_H), \tag{30}$$

which says that bankruptcy costs are equal to $1-\lambda_H$ if the number of bankruptcies $F(\hat{v})$ is below the threshold "court capacity" κ and increase to $1 - \lambda_L$ if they are above it.

From Proposition 2, we know that high bankruptcy costs impede restructuring, making it hard to reduce debt, and hence making bankruptcy itself more likely. Now, with congestion costs, this can create an amplification spiral. If bankruptcy filings exceed the court's threshold ($F(\hat{v}) >$

κ), bankruptcy costs increase, by assumption. This reduces the feasibility of restructuring (by Proposition 2) and increases filings, closing the loop.

The spiral has the potential to generate financial instability in the form of multiple equilibria:

Proposition 10. *Suppose*

$$\hat{v}\big(D^*_{\lambda=\lambda_H}\big) < F^{-1}(\kappa) < \hat{v}\big(D^*_{\lambda=\lambda_L}\big), \tag{31}$$

where D^*_λ is the face value that makes creditors' IC (inequality (4)) bind in the baseline model for the indicated value of λ. There are two equilibria:

- *There is a "good" equilibrium in which the probability of filing is low, courts are not congested, and the costs of bankruptcy are low; and*
- *there is a "bad" equilibrium in which the probability of filing is high, courts are congested, and the costs of bankruptcy are high.*

This result suggests that bankruptcy policy cannot be separated from financial stability regulation: Congestion itself can create panic-like coordination failures. Bankruptcy policy is not just about mitigating the costs of filings at the margin, but also about preventing mass filings altogether. Indeed, increasing court capacity κ—so that the second inequality in condition (31) is violated—can eliminate the "bad" equilibrium. This adds support to the argument that avoiding court congestion should be a policy priority in times of crisis (see Iverson, Ellias, and Roe (2020)).

D.2 Endogenous Asset Values and Debt Overhang

We have assumed thus far that, although the asset value v is uncertain ex ante, its ex post distribution is exogenous. We have therefore focused on so-called "direct" costs of liquidation and bankruptcy, ignoring the "indirect" costs that can arise due to, e.g., debt overhang (Myers (1977)). Here, we incorporate endogenous asset values and show conditions under which debt overhang amplifies or attenuates our results.

We assume that the firm can make an investment before the asset value v is realized, exerting effort η to improve the distribution of v. Specifically, we assume that increasing η to $\eta' > \eta$ improves the distribution of v from F^η to $F^{\eta'} \succ F^\eta$, where "$\succ$" indicates first-order stochastic dominance. We also assume that the firm will exert less effort when it has high levels of debt because the costs of effort are borne solely by equity holders, whereas the benefits are shared with creditors. (Rather than model the firm's decision directly, however, we simply assume the firm exerts more effort when it has less debt.)

To formulate a comparative-statics result, we define the parameter λ^η as follows: the smallest λ for which it is feasible to write down debt sufficiently to reduce the bankruptcy filing threshold, $\hat{v}$, to a given level.[63,64] The next result describes how λ^η depends on η.

Proposition 11. *If the probability of default $F(\hat{v})$ does not depend on η, then λ^η is decreasing in η. On the other hand, if the tail conditional expectation $\mathbb{E}[v \mid v \leq \hat{v}(D)]$ does not depend on η, then λ^η is increasing in η.*

This result reveals that the effect of debt overhang on restructuring depends on how the overhang affects the distribution of values ex post. High effort can have two effects, one present under each of the conditions in the proposition.

- If effort does not affect the probability of default $F(\hat{v})$, then increasing η makes restructuring easier. In this situation, higher effort increases the probability of high asset values, increasing creditor recoveries in bankruptcy and making priority more valuable. In this case, the more there is to gain from restructuring, the more likely it is to occur—creditors restructure to access these gains.

- If, on the other hand, effort affects the probability of default, $F(\hat{v})$, but does not affect the value of the firm's assets in bankruptcy $\mathbb{E}[v|v \leq \hat{v}]$, then increasing η makes restructuring harder. In this situation, higher effort reduces the probability of bankruptcy, reducing the

[63] We work with $\hat{v}$ (inequality (3)) instead of the debt D directly only because it makes the math easier.

[64] Proposition 2, which shows that high λ facilitates restructuring, implies such a λ^η is well-defined.

value of seniority and making priority less valuable. In this case, the more there is to gain from restructuring, the less likely it is to occur—creditors hold out to free ride on these gains.

Overall, this result adds nuance to Brunnermeier and Krishnamurthy's (2020) argument that an efficient bankruptcy system helps resolve debt-overhang problems.

D.3 Ex Ante Borrowing and Debt Capacity

So far, we have taken the initial debt level D_0 as exogenous. Our results are thus exposed to a Lucas-type critique: By studying the effect of the bankruptcy environment on how the debt is restructured, we neglect to study its effect on how the debt comes to be in the first place. This approach has strengths: With D_0 as a starting point, not an outcome, our analysis does not depend on its source. By contrast, endogenizing D_0 could require specific assumptions about the firm's motives for borrowing, such as exploiting tax shields as in the trade-off theory, financing investment as in q theory, or disciplining management as in agency theory. Our model also retains tractability while allowing for general asset distributions (F), whereas models with endogenous debt, by and large, do not.

Nonetheless, we analyze the firm's ex ante borrowing here. We develop a specific model of endogenous debt. We show that our main comparative statics results—how it affects the write-down $D_0 - D$ (Proposition 2 and Proposition 3)—are robust to endogenizing D_0, albeit only via numerical examples. In the main text (see Proposition 6), we analyze policy taking unique capacity itself as the objective. We show that our policy conclusions coincide with those in Section 5, in which the objective is ex post utilitarian welfare.

The starting point for the analysis is creditors' ex ante expected repayment, which we denote by L. An individual creditor with (unsecured) debt with face value D_0 anticipates a restructuring, in which all other creditors restructure to (secured) debt with total face value D, making the value of its initial (unsecured) debt

$$L = \big(1 - F(\hat{v}(D)\big)D_0. \tag{32}$$

Here we suppose that the firm takes debt D_0 to borrow a fixed amount I from a continuum of competitive creditors, presumably to finance an investment. Net of the investment, the firm value is v, which we suppose here is uniform: $v \sim U[0, \bar{v}]$. Everything else is as described in Section 2.

The face value D is determined by creditors' break-even condition. Given that, in a symmetric equilibrium, all creditors start with the same debt D_0, which is restructured to D. So, given their ICs bind in a restructuring, their expected repayment is as if they borrowed with secured debt with face value D initially and the creditors' break-even condition, $L = I$ (cf. equation (32)) reads

$$\big(1 - F(\hat{v}(D))\big)D + F(\hat{v}(D))\mathbb{E}[\lambda\theta v \,|\, v \leq \hat{v}(D)] = I. \tag{33}$$

This depends on D_0 via the face value D, which, in turn, is the outcome of a restructuring per equation (4). Substituting $F(v) = v/\bar{v}$ and $D = (1 - (1-\theta)\lambda)\hat{v}$, this can be rewritten as

$$\left(1 - \frac{\hat{v}}{\bar{v}}\right)\big(1 - (1-\theta)\lambda\big)\hat{v} + \frac{\lambda\theta\hat{v}^2}{2\bar{v}} = I. \tag{34}$$

This is a quadratic equation in $\hat{v}$ with a unique positive solution

$$\hat{v} = \frac{(1-\theta)\lambda - 1 + \sqrt{\big(1 - (1-\theta)\lambda\big)^2 - \frac{4I}{\bar{v}}\left(\frac{\lambda\theta}{2} - \big(1 - (1-\theta)\lambda\big)\right)}}{\frac{2}{\bar{v}}\left(\frac{\lambda\theta}{2} - \big(1 - (1-\theta)\lambda\big)\right)}. \tag{35}$$

We can combine this expression for $\hat{v}$ from the break-even condition with the expression for the write-down (Proposition 1) to analyze how the write-down depends on λ and θ.

The restructuring condition gives an expression for the write-down $D_0 - D$ in terms of $\hat{v}$:

$$D_0 - D = \frac{F(\hat{v}(D))}{1 - F(\hat{v}(D))}\mathbb{E}[\lambda\theta v \,|\, v \leq \hat{v}(D)] = \frac{\lambda\theta\hat{v}_D^2}{2(\bar{v} - \hat{v}_D)}, \tag{36}$$

having made use of the assumption that v is uniform. Substituting the expression for $\hat{v}$ from the break-even condition (3), we can plot the write-down as a function of λ and θ.

Figure 4 illustrates our main results in this model with endogenous D_0: The equilibrium write-down is generally increasing in bankruptcy efficiency λ (Proposition 2) and can be increasing, decreasing, or hump-shaped in creditor friendliness θ (Proposition 3).

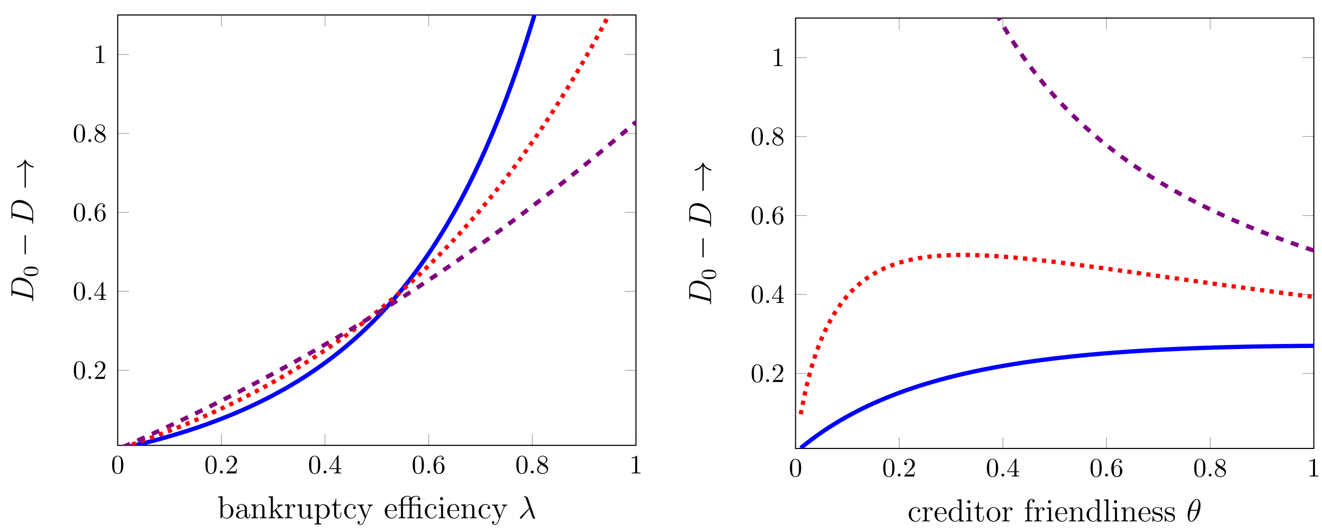

Figure 4: The plots above show how the equilibrium write-down $D_0 - D$ depends on λ (left panel) and θ (right panel) with D_0 endogenous. The parameters used are as follows: both panels: $F(v) = v/100$ and $I = 10$; left panel: $\theta = 0.4$ (solid blue), 0.6 (dotted red), 0.8 (dashed violet); right panel: $\lambda = 0.4$ (solid blue), 0.6 (dotted red), 0.8 (dashed violet).

D.4 Tort Claimants

Here, we study the role of "accidental"/tort creditors. To do so, we suppose the firm has outstanding tort claims equal to T. If T is not paid in full prior to a bankruptcy filing, different priority rules correspond to different types of taxes in bankruptcy: If tort claims are treated on-par with secured claims, they are equivalent to tax τ_s on senior debt. If they are treated on-par with unsecured debt, they represent a tax τ_j on junior debt. If they are junior to unsecured debt, there is no tax on creditors (and they will often go unpaid).

Within this set-up, we now return to creditors' incentive to accept a restructuring. Their IC becomes

$$\begin{aligned}&\big(1 - F(\hat{v}(D+T))\big)D + F\big(\hat{v}(D+T)\big)\mathbb{E}\left[(1-\tau_s)\lambda\theta v \,\middle|\, v < \hat{v}(D+T)\right] \geq \\ &\big(1 - F(\hat{v}(D+T))\big)D_0 + (1-\rho)F(\hat{v}(D+T))\mathbb{E}\left[(1-\tau_j)\lambda\theta v \,\middle|\, v < \hat{v}(D+T)\right]\frac{D_0}{D}.\end{aligned} \tag{37}$$

This condition is easiest to satisfy if τ_s is small and τ_j is large. This suggests that, to facilitate restructuring, tort claimants should be paid behind secured debt, but ahead of unsecured debt. That ordering makes priority valuable by (i) increasing the value of secured debt and (ii) decreasing the value of unsecured debt.

E Mixed Offers

In our baseline model, we study exchange offers that include a single security, equity or debt. In Section 6.2, we show that offers that include a mix and debt and equity could arise with a concentrated creditor. Here, we show they never arise in our baseline model with dispersed creditors.

Suppose that the firm offers creditors a mix of a proportion of equity $1-\varphi$ and senior debt with face value D in exchange for their junior debt D_0.

Noting that the bankruptcy decision condition in equation (3) is unchanged by new equity (i.e. that the firm will file whenever $v \leq \hat{v}(D)$), we can write a creditor's payoffs as follows:

- If it accepts, it gets

$$\begin{aligned}\text{payoff}\,\Big|_{\text{acc.}} = \big(1-F(\hat{v})\big)D + \int_0^{\hat{v}}(\theta + (1-\varphi)(1-\theta))\lambda v \mathrm{d}F(v) \\ + \int_{\hat{v}}^{\infty}(1-\varphi)(v-D)\mathrm{d}F(v).\end{aligned} \tag{38}$$

- If it rejects, it gets

$$\text{payoff}\big|_{\text{rej.}} = \big(1 - F(\hat{v})\big)D_0. \tag{39}$$

So the firm chooses φ and D to

$$\text{maximize} \int_0^{\hat{v}} \varphi\lambda(1-\theta)vf(v)\mathrm{d}v + \int_{\hat{v}}^{\infty} \varphi(v-D)f(v)\mathrm{d}v \tag{40}$$

subject to creditors' IC that

$$\text{payoff}\big|_{\text{acc.}} \geq \text{payoff}\big|_{\text{rej.}} \tag{41}$$

Supposing that the constraint binds and substituting it in the objective, we can rewrite the problem as

$$\text{maximize} \int_0^{\hat{v}} \lambda v\mathrm{d}F(v) + \int_{\hat{v}}^{\bar{v}} v\mathrm{d}F(v) - (1 - F(\hat{v}(D)))D_0. \tag{42}$$

Now observe that this does not depend on φ and that D appears only in $\hat{v}$. Hence, if there is an interior optimum, we can maximize the objective with respect to $\hat{v}$ directly to get

$$-\lambda f(\hat{v})\hat{v} + f(\hat{v})\hat{v} + f(\hat{v})D_0 = 0. \tag{43}$$

Substituting in for $\hat{v}$ from equation (3) and solving, we find that:

$$D = \frac{1 - \lambda(1-\theta)}{1-\lambda}D_0 > D_0. \tag{44}$$

Still supposing an interior optimum, substitute D_0 into the binding constraint in (41) to get that

$$\begin{aligned}(1-\varphi)\left(\int_0^{\hat{v}} \Delta + (1-\theta)\lambda v\mathrm{d}F(v) + \int_{\hat{v}}^{\infty} (v-D)\mathrm{d}F(v)\right) &= \\ = -\big(1 - F(\hat{v})\big)\big)\lambda\theta\hat{v}(D) - \int_0^{\hat{v}} \lambda\theta v\mathrm{d}F(v).\end{aligned} \tag{45}$$

But this implies that new equity is negative. So we conclude that there is not an interior optimum, but, rather, that $1-\varphi=0$. Substituting into the constraint (41), we get

$$(1-F(\hat{v}))D+\int_0^{\hat{v}}\lambda\theta v f(v)\mathrm{d}v \geq (1-F(\hat{v}))D_0, \tag{46}$$

which is the usual constraint.

F Proofs

F.1 Proof of Lemma 1

Creditors are better off accepting the equity share $1-\varphi$ if its value is greater than their expected payoff in bankruptcy:

$$(1-\varphi)\mathbb{E}[v] \geq \mathbb{E}\big[\mathbb{1}_{\{v\geq\hat{v}\}}D_0+\mathbb{1}_{\{v<\hat{v}\}}\lambda\theta v\big], \tag{47}$$

where $\hat{v}$ here denotes $\hat{v}(D_0)\equiv D_0/\big(1-(1-\theta)\lambda\big)$. Similarly, equity holders are better off if their residual claim (φ of the assets) is worth more than what they expect in bankruptcy:

$$\varphi\mathbb{E}[v] > \mathbb{E}\big[\mathbb{1}_{\{v\geq\hat{v}\}}(v-D_0)+\mathbb{1}_{\{v<\hat{v}\}}(1-\theta)\lambda v\big] \equiv \mathbb{E}\big[\max\{v-D_0,(1-\theta)\lambda v\}\big]. \tag{48}$$

These inequalities can be rewritten and combined as

$$\frac{\mathbb{E}\big[v-\mathbb{1}_{\{v\geq\hat{v}\}}D_0-\mathbb{1}_{\{v<\hat{v}\}}\lambda\theta v\big]}{\mathbb{E}[v]} > \varphi > \frac{\mathbb{E}\big[v-\mathbb{1}_{\{v\geq\hat{v}\}}D_0-\mathbb{1}_{\{v<\hat{v}\}}\lambda\theta v\big]-(1-\lambda)\mathbb{E}\big[\mathbb{1}_{\{v<\hat{v}\}}v\big]}{\mathbb{E}[v]}, \tag{49}$$

Since the left-most term is always strictly greater than the right-most term, an appropriate debt-to-equity restructuring can implement a strict Pareto improvement (and avoid all costs of financial distress). □

F.2 Proof of Lemma 2

First, recall that each creditor accepts a restructuring offer only if it makes the creditor better off, given that other creditors accept. That is, an individual creditor must prefer getting a fraction $1-\varphi$ of the assets to holding its original debt with face value D_0. If all other creditors agree to the restructuring, the firm is effectively all equity (assuming the individual creditor is infinitesimally small). A creditor therefore accepts if

$$(1-\varphi)\mathbb{E}[v] \geq D_0. \tag{50}$$

Similarly, equity holders are better off in a restructuring if their residual claim on the fraction φ of the assets is worth more than their bankruptcy payoff, as in inequality (48). These inequalities can be rewritten and combined as

$$\mathbb{E}[v] - D_0 \geq \varphi\mathbb{E}[v] \geq \mathbb{E}[v] - D_0 + \mathbb{E}\big[\mathbb{1}_{\{v<\hat{v}\}}\big\{(1-\theta)\lambda v - (v - D_0)\big\}\big]. \tag{51}$$

The last expectation is positive, because the term in braces is positive for $v < \hat{v}$ by the definition of $\hat{v}$ (equation (3)). Hence, the right-most term is greater than the left-most term; no restructuring of debt to equity is feasible. $\square$

F.3 Proof of Lemma 3

An individual creditor will accept a restructuring to pari passu debt if

$$\begin{aligned}\Big(1-F\big(\hat{v}(D)\big)\Big)D + F\big(\hat{v}(D)\big)\mathbb{E}\left[\lambda\theta v \,\middle|\, v < \hat{v}(D)\right] \geq \\ \geq \Big(1-F\big(\hat{v}(D)\big)\Big)D_0 + F\big(\hat{v}(D)\big)\frac{D_0}{D}\mathbb{E}\left[\lambda\theta v \,\middle|\, v < \hat{v}(D)\right].\end{aligned} \tag{52}$$

The left-hand side of the inequality above is the expected payoff if the creditor accepts: If there is no future bankruptcy, the creditor gets D; if there is one, creditor gets a unit share of the recovery (given it has the same face value D as the unit of other identical creditors). The right-hand side is the expected payoff if the creditor holds out: If there is no future bankruptcy, it gets D_0; if there is, it gets a share D_0/D of the recovery value (given it has face value D_0 and each of the unit of other identical creditors have face value D). Observe that, because the creditor is small, its decision does not affect the firm's filing decision; hence, the bankruptcy threshold $\hat{v}$ depends on the new level of total debt D even if the creditor holds out. Denoting the price of (unsecured) debt with face value one by e^{-y^u} (y^u is the continuously compounded yield to maturity), this condition can be rewritten as

$$De^{-y^u} \geq D_0 e^{-y^u}. \tag{53}$$

This is never satisfied, implying immediately that no restructuring to equal priority debt is feasible.

F.4 Proof of Proposition 1

The argument for the first part of the result is in the text.

To show that accepting is a dominant strategy, we denote the proportion of creditors accepting the offer by ψ. An individual creditor takes as given the aggregate face value: $\psi D + (1-\psi)D_0$. Thus, the corresponding difference in payoffs is

$$\Delta_\psi := \underbrace{\Big(1 - F\big(\hat{v}(\psi D + (1-\psi)D_0)\big)\Big)D + F\big(\hat{v}(\psi D + (1-\psi)D_0)\big)\mathbb{E}\left[\lambda\theta v \,\middle|\, v < \hat{v}(\psi D + (1-\psi)D_0)\right]}_{\text{payoff if accepts}} - \underbrace{\Big(1 - F\big(\hat{v}(\psi D + (1-\psi)D_0)\big)\Big)D_0}_{\text{payoff if holds out}}. \tag{54}$$

Differentiating w.r.t. ψ, we get

$$\frac{\partial\Delta}{\partial\psi} = (D_0 - D + \lambda\theta)\, f(\hat{v})\hat{v}\frac{\partial\hat{v}\big(\psi D + (1-\psi)D_0\big)}{\partial\psi} = -\,(D_0 - D + \lambda\theta)\,\frac{(D_0 - D)}{1-(1-\theta)\lambda} f(\hat{v})\hat{v} < 0, \tag{55}$$

i.e. accepting is less attractive when more other creditors accept (ψ is higher). Therefore if creditors accept for $\psi = 1$ they accept for $\psi < 1$, i.e. accepting is a dominant strategy. □

F.5 Proof of Proposition 2

First, we rewrite Δ in equation (4) as

$$\Delta := \Big(1 - F\big(\hat{v}(D)\big)\Big)\big(D - D_0\big) + \lambda\theta\int_0^{\hat{v}(D)} v\mathrm{d}F(v). \tag{56}$$

The maximum write-down $D_0 - D^*$ corresponds to the minimum face value D^*, or to $\Delta = 0$. To see how D^* depends on λ, we use the chain rule to write:

$$\frac{\partial D^*}{\partial\lambda} = -\frac{\partial\Delta/\partial\lambda}{\partial\Delta/\partial D}. \tag{57}$$

Computing, we see that the denominator is positive:

$$\frac{\partial\Delta}{\partial D} = f(\hat{v})\left(\frac{\partial\hat{v}}{\partial D}D_0 - \hat{v}\right) + 1 - F(\hat{v}) + \lambda\theta\hat{v}f(\hat{v})\frac{\partial\hat{v}}{\partial D} > 0, \tag{58}$$

given that all terms are positive.[65] The numerator is positive too:

$$\frac{\partial\Delta}{\partial\lambda} = \theta\int_0^{\hat{v}(D^*)} v\mathrm{d}F(v) + \Big(\lambda\theta\hat{v} + D_0 - D^*\Big)f(\hat{v})\frac{\partial\hat{v}}{\partial\lambda} > 0. \tag{59}$$

This proves the result in the text. □

[65] To see why, note that

$$\frac{\partial\hat{v}}{\partial D}(D)D_0 = \hat{v}(D_0)$$

and $D_0 > D^*$.

F.6 Proof of Proposition 3

To prove the first part of the result—how D^* depends on θ—we use the chain rule to write

$$\frac{\partial D^*}{\partial \theta} = -\frac{\partial \Delta / \partial \theta}{\partial \Delta / \partial D}. \tag{60}$$

The denominator is as in equation (58) above. Recall that it is positive. The numerator is

$$\frac{\partial \Delta}{\partial \theta} = \lambda \int_0^{\hat{v}} v \mathrm{d}F(v) + \big(\lambda \theta \hat{v} + D_0 - D^*\big) f(\hat{v}) \frac{\partial \hat{v}}{\partial \theta}. \tag{61}$$

Given that the last term is negative, that is,

$$\frac{\partial \hat{v}}{\partial \theta} = -\frac{\lambda D}{\big(1-(1-\theta)\lambda\big)^2} < 0, \tag{62}$$

this expression can change sign depending on parameters. This proves the first part of the result.

To prove the second part of the result on the existence of an interior θ^*, we show that it follows from continuity: We show that θ^* is always decreasing in θ at $\theta = 0$ and, under the condition in the result, is increasing in θ at $\theta = 1$, so D^* must be minimized for an interior value $\theta^* \in (0, 1)$.

From equations (58) and (60), we know that $\partial D^* / \partial \theta$ has the opposite sign of $\partial \Delta / \partial \theta$, which is given in equation (61). We compute:

- At $\theta = 0$, we have $\Delta = (1 - F(\hat{v}))(D - D_0)$; hence, $D^* = D_0$. Now, substituting into equation (61),

$$\left.\frac{\partial \Delta}{\partial \theta}\right|_{\theta=0} = \lambda \int_0^{\hat{v}} v \mathrm{d}F(v) + \big(D_0 - D^*\big) f(\hat{v}) \frac{\partial \hat{v}}{\partial \theta} \tag{63}$$

$$= \lambda \int_0^{\hat{v}} v \mathrm{d}F(v) > 0. \tag{64}$$

- At $\theta = 1$, we have $\hat{v} = D$ and $\Delta = (1-F(D))(D-D_0)+\lambda \int_0^D v dF(v)$; hence, $\lambda \int_0^{D^*} v dF(v) =$

$(1-F(D^*))(D_0-D^*)$. Now, substituting into equation (61),

$$\left.\frac{\partial\Delta}{\partial\theta}\right|_{\theta=1} = \lambda\int_0^{D^*} v\mathrm{d}F(v) - (D_0-(1-\lambda)D^*)\lambda D^* f(D^*) \tag{65}$$

$$= (D_0-D^*)\Big[(1-F(D^*)) - \lambda D^* f(D^*)\Big] - \lambda^2(D^*)^2 f(D^*). \tag{66}$$

Given $D_0 > D^*$, a sufficient condition for this to be negative (and hence for the existence of an interior minimum) is for the term in square brackets to be negative (at $\theta = 1$), which is the condition in the statement of the result. □

F.7 Proof of Proposition 4

We have to work with equation (12). Given that the denominator $\partial\Delta/\partial\hat{v}$ is the same for each subsidy and we know the bang for the buck q_i, we need only to calculate the numerator $-\partial\Delta/\partial s_i$ for each subsidy in turn. Using the subsidy-adjusted filing threshold

$$\hat{v}(\mathbf{s}) = \frac{D + s_E^B - s_E^O}{1-\lambda(1-\theta)} \tag{67}$$

to Δ as

$$\Delta = \int_0^{\hat{v}} \Big(\lambda\theta v + s_S^B - s_U^B\Big)\mathrm{d}F(v) + \int_{\hat{v}}^{\infty} \Big((1-\lambda(1-\theta))\hat{v} + s_E^O - s_E^B - D_0 + s_S^O - s_U^O\Big)\mathrm{d}F(v), \tag{68}$$

we derive the sensitivities $-\partial\Delta/\partial s_i$ for each s_i as follows:

1. Equity inside bankruptcy: $-\partial\Delta/\partial s_E^B = 1 - F(\hat{v})$.

2. Unsecured debt inside bankruptcy: $-\partial\Delta/\partial s_U^B = F(\hat{v})$.

3. Secured debt inside bankruptcy: $-\partial\Delta/\partial s_S^B = -F(\hat{v})$.

4. Equity outside bankruptcy: $-\partial\Delta/\partial s_E^O = -(1 - F(\hat{v}))$.

5. Unsecured debt outside bankruptcy: $-\partial\Delta/\partial s_U^O = 1 - F(\hat{v})$.

6. Secured debt outside bankruptcy: $-\partial\Delta/\partial s_S^O = -(1 - F(\hat{v}))$.

Multiplying these sensitivities by the "bang for the buck" $1/q_i$, with $q_i = F(\hat{v})$ for subsidies inside bankruptcy and $1 - F(\hat{v})$ for outside, equation (12) gives the expressions in the table (Table I).

Finally, to ensure that we want to choose the smallest entry in Table I, i.e. that minimizing $\mathrm{d}\hat{v}/\mathrm{d}\varepsilon$ is equivalent to minimizing $-(\partial\Delta v/\partial s_i)/q_i$, we need to check that $\partial\Delta/\partial\hat{v} > 0$ at $\mathbf{s} = \mathbf{0}$ in equation (12). This holds:

$$\begin{aligned} \left.\frac{\partial\Delta}{\partial\hat{v}}\right|_{\mathbf{s}=0} =& [\lambda\theta\hat{v} + s_S^B - s_U^B - (1-\lambda(1-\theta))\hat{v} \\ & \left. - s_E^O + s_E^B + D_0 - s_S^O + s_U^O]f(\hat{v}) + (1-\lambda(1-\theta))(1-F(\hat{v}))\right|_{\mathbf{s}=0} \end{aligned} \tag{69}$$

$$=(D_0 - (1-\lambda)\hat{v})f(\hat{v}) + (1-\lambda(1-\theta))(1-F(\hat{v})) > 0. \tag{70}$$

$\square$

F.8 Proof of Proposition 5

Proposition 4 gives the optimal policy for $\lambda \lesseqgtr 1$. Here we show that the menu in the proposition leads creditors to self-select into the right policy. If creditors choose a subsidy to themselves inside bankruptcy, given a price p per unit subsidy $q_S^B s_S^B$, they get

$$v_D(\mathbf{s}) = (1 - F(\hat{v}(\mathbf{s})))(D_0 + s_U^O) - p\, q_S^B s_S^B. \tag{71}$$

To find creditors' preferred (marginal) policy, we differentiate w.r.t. the subsidies to get, via equation (12), for the subsidy $s_i \in \{s_E^O, s_E^B, s_S^O, s_S^B\}$:

$$\frac{\mathrm{d}v_D(\mathbf{s})}{\mathrm{d}\varepsilon}\Big|_{\varepsilon=0} = \frac{1}{q_i}\frac{\mathrm{d}v_D(\mathbf{s})}{\mathrm{d}s_i}\Big|_{\varepsilon=0} \tag{72}$$

$$= -f(\hat{v})\frac{D_0}{q_i}\frac{\mathrm{d}\hat{v}(\mathbf{s})}{\mathrm{d}s_i}\Big|_{\varepsilon=0} - p\mathbb{1}_{\{s_i=s_S^B\}} \tag{73}$$

$$= \begin{cases} \dfrac{f(\hat{v})D_0}{\partial\Delta/\partial\hat{v}}\Big|_{\varepsilon=0} - p\mathbb{1}_{\{s_i=s_S^B\}} & \text{if } s_i \in \{s_E^O, s_S^O, s_S^B\}, \\ \\ -\dfrac{1-F(\hat{v})}{F(\hat{v})}\dfrac{f(\hat{v})D_0}{\partial\Delta/\partial\hat{v}}\Big|_{\varepsilon=0} & \text{if } s_i = s_E^B \end{cases} \tag{74}$$

having used the value of $\dfrac{\mathrm{d}\hat{v}(\mathbf{s})}{\mathrm{d}s_i}$ from the proof of Proposition 4.

Creditors choose the welfare-maximizing subsidy if the payoff of choosing s_E^B exceeds that of choosing one of the others if and only if $\lambda > 1$:

$$\frac{f(\hat{v})D_0}{\partial\Delta/\partial\hat{v}} - p < -\frac{1-F(\hat{v})}{F(\hat{v})}\frac{f(\hat{v})D_0}{\partial\Delta/\partial\hat{v}} \iff \lambda > 1. \tag{75}$$

It holds if and only if

$$p = \frac{f(\hat{v})}{F(\hat{v})}\frac{D_0}{\partial\Delta/\partial\hat{v}}\Big|_{\lambda=1,\varepsilon=0} = \frac{1}{F(\hat{v})[1+\frac{1-F(\hat{v})}{\hat{v}f(\hat{v})}\frac{D^*}{D_0}]}\Big|_{\lambda=1,\varepsilon=0} = \frac{1}{\pi(1)} \tag{76}$$

and

$$\sup_{\lambda\in(0,1)} \pi(\lambda) \le \pi(1) \le \inf_{\lambda\in(0,1/(1-\theta))} \pi(\lambda), \tag{77}$$

namely if the conditions of the proposition are satisfied. □

F.9 Proof of Proposition 6

To assess the marginal effects of subsidies, we differentiate L in equation (15):

$$\frac{\mathrm{d}L(\mathbf{s})}{\mathrm{d}s_S^B} = F(\hat{v}(\mathbf{s})); \ \frac{\mathrm{d}L(\mathbf{s})}{\mathrm{d}s_S^O} = 1 - F(\hat{v}(\mathbf{s})); \ \frac{\mathrm{d}L(\mathbf{s})}{\mathrm{d}s_E^B} = -(1 - F(\hat{v}(\mathbf{s}))); \ \frac{\mathrm{d}L(\mathbf{s})}{\mathrm{d}s_E^O} = 1 - F(\hat{v}(\mathbf{s})), \quad (78)$$

where we have disregarded the effects of subsidies on L via $\hat{v}$, which are negligible due to the optimality of $\hat{v}$ (by the envelope theorem). Dividing by the prices $\mathbf{q}$ from Section 5 and taking the limit as $\varepsilon \to 0$ gives the marginal impact of each policy on L:

$$\frac{1}{q_S^B}\frac{\mathrm{d}L(\mathbf{s})}{\mathrm{d}s_S^B} = 1; \ \frac{1}{q_S^O}\frac{\mathrm{d}L(\mathbf{s})}{\mathrm{d}s_S^O} = 1; \ \frac{1}{q_E^B}\frac{\mathrm{d}L(\mathbf{s})}{\mathrm{d}s_E^B} = -\frac{1 - F(\hat{v})}{F(\hat{v})}; \ \frac{1}{q_E^O}\frac{\mathrm{d}L(\mathbf{s})}{\mathrm{d}s_E^O} = 1. \quad (79)$$

The ranking in the proposition follows. □

When $\lambda > 1$, the debt capacity is strictly increasing in $\hat{v}$ in a neighborhood of $\mathbf{s} = \mathbf{0}$, so the firm takes debt with an infinite face value and always defaults. The debt capacity is $L(\mathbf{s}) = \lambda\theta\mathbb{E}[v] + s_S^B$, and the unique optimal policy is to subsidize creditors in bankruptcy.

F.10 Proof of Corollary 1

The result follows immediately from the expression for debt capacity under subsidies (equation 79). □

F.11 Proof of Lemma 6

First, define the difference in a creditor's payoffs from accepting versus rejecting an offer, given that other creditors accept:

$$\Delta = (1 - F(\hat{v}(D_2)))(D_2 - D_1) + F(\hat{v}(D_2))\mathbb{E}[\rho\lambda\theta v | v < \hat{v}(D_2)] \quad (80)$$

$$= (1 - F(\hat{v}(D_2)))(D_2 - D_1) + \rho\int_0^{\hat{v}(D_2)} \lambda\theta v \mathrm{d}F(v). \quad (81)$$

Now compute the derivative $\mathrm{d}D_2^*/\mathrm{d}\rho$ in two steps:

$$\frac{\partial\Delta}{\partial D_2} = (1 - F(\hat{v}(D_2)) + (D_1 - D_2 + \rho\lambda\theta\hat{v}(D_2))\, f(\hat{v}(D_2))\frac{\partial\hat{v}(D_2)}{\partial D_2} > 0 \tag{82}$$

and

$$\frac{\partial\Delta}{\partial\rho} = \int_0^{\hat{v}(D_2)} \lambda\theta v \mathrm{d}F(v) > 0. \tag{83}$$

So,

$$\frac{\mathrm{d}D_2^*}{\mathrm{d}\rho} = -\frac{\partial\Delta/\partial\rho}{\partial\Delta/\partial D} < 0 \tag{84}$$

and $\rho = 1$ maximizes the write-down. □

F.12 Proof of Proposition 7 and Corollary 2

To start, let $\Delta_1 := (1 - F(\hat{v}(D_2)))D_1 + (1-\rho)F(\hat{v}(D_2))\mathbb{E}[\lambda\theta v|v < \hat{v}(D_2)] - (1 - F(\hat{v}(D_2))D_0$. be the wedge between accepting D_1 and holding on to D_0. Now apply the implicit function theorem to $\Delta_1 = 0$ keeping in mind that both D_0 and D_2 do not depend on ρ, to get

$$\frac{\mathrm{d}D_1}{\mathrm{d}\rho} = -\frac{\partial\Delta_1/\partial\rho}{\partial\Delta_1/\partial D_1} = \frac{F(\hat{v}(D_2))\mathbb{E}[\lambda\theta v|v < \hat{v}(D_2)]}{1 - F(\hat{v}(D_2))} > 0. \tag{85}$$

Thus, $D_0 - D_1$ is decreasing in ρ. When $\rho = 0$, $D_2 = D_1$ solves $\Delta_1 = 0$; when $\rho = 1$, $D_1 = D_0$ solves the first-round IC $(1 - F(\hat{v}(D_2)))D_1 = (1 - F(\hat{v}(D_2))D_0$. □

F.13 Proof of Proposition 8

Under the new legal remedy, the second-round IC is

$$\begin{aligned}
&\big(1 - F(\hat{v}(D_2))\big)D_2^\phi + F(\hat{v}(D_2))\mathbb{E}\left[\frac{\lambda\theta v}{\phi} - (1-\rho)\frac{(1-\phi)\tau(v)}{\phi} \mid v < \hat{v}(D_2)\right] \\
&\geq \big(1 - F(\hat{v}(D_2))\big)D_1 + (1-\rho)\, F(\hat{v}(D_2))\mathbb{E}[\tau(v) \mid v < \hat{v}(D_2)]\,.
\end{aligned} \tag{86}$$

which, after substituting D_2^{ϕ} in, is reduced to

$$D_1 - D_2 \quad \leq \frac{F(\hat{v}(D_2))}{\big(1 - F(\hat{v}(D_2))\big)} \mathbb{E}[\lambda\theta v - (1-\rho)\tau(v) \mid v < \hat{v}(D_2)] \,. \tag{87}$$

Clearly, the fraction ϕ is irrelevant for the equilibrium outcome D_2. Creditors' first-round IC is therefore

$$\big(1 - F(\hat{v}(D_2))\big) D_1 + (1-\rho)\, F(\hat{v}(D_2))\mathbb{E}[\tau(v) \mid v < \hat{v}(D_2)] \geq (1 - F(\hat{v}(D_2)) D_0 \tag{88}$$

Adding up the first round IC for $\phi = 1$, given the irrelevance of ϕ, and the second round IC yields the same IC as the case when the transaction was reverted with probability $1-\rho$, or when there was never a second round to start with. So the equilibrium outcome D_2, and therefore the expected payoff of everyone, are independent of the specific legal remedies in court.

To see the debt dynamics, let $\Delta_1 := \big(1-F(\hat{v}(D_2))\big) D_1 + (1-\rho)\, F(\hat{v}(D_2))\mathbb{E}[\tau(v) \mid v < \hat{v}(D_2)] - (1 - F(\hat{v}(D_2)) D_0$ be the wedge between accepting D_1 and holding on to D_0. Now apply the implicit function theorem to $\Delta_1 = 0$ keeping in mind that both D_0 and D_2 do not depend on ρ, to get

$$\frac{\mathrm{d}D_1}{\mathrm{d}\rho} = -\frac{\partial\Delta_1/\partial\rho}{\partial\Delta_1/\partial D_1} = \frac{F(\hat{v}(D_2))\mathbb{E}[\tau(v) \mid v < \hat{v}(D_2)]}{1 - F(\hat{v}(D_2))} > 0. \tag{89}$$

Thus, $D_0 - D_1$ is decreasing in ρ. When $\rho = 0$, $D_2 = D_1$ solves $\Delta_1 = 0$; when $\rho = 1$, $D_1 = D_0$ solves the first-round IC $(1 - F(\hat{v}(D_2)))D_1 = (1 - F(\hat{v}(D_2))D_0$. □

F.14 Proof of Proposition 9

Here, we solve for the restructuring offer that equity holders make if they face a large, concentrated creditor (denoted by L below) with probability ξ and small, dispersed creditors (denoted by S) with probability $1-\xi$.

Ultimately, there are three possibilities:

1. The firm makes an offer that L accepts and S reject.

2. The firm makes an offer the L rejects and S accept.

3. The firm makes an offer that both L and S accept.

The proof involves simply (i) computing current equity's optimal restructuring given each of these possibilities and (ii) comparing its payoffs (denoted by u below) from each possibility given these optimal restructurings. But this involves some steps:

Step 1: Write L's and S's ICs to accept a restructuring offer.

Step 2: Write current equity's payoffs u in benchmarks in which $\xi = 1$ and $\xi = 0$. That is, creditors are concentrated for sure or dispersed for sure (these expressions are useful in the subsequent comparisons).

Step 3: Calculate current equity's payoffs u for each possibility 1–3 above.

Step 4: Compare these payoffs to determine the optimal restructuring offer.

Step 1: IC constraints. Here, we define the conditions for each type of creditor to accept the firm's offer of a restructuring to debt D and equity $1 - \varphi$.

- *Large creditor's IC.* Define the difference in its payoffs from accepting and rejecting the offer (inequality (20) with the expectations expanded as integrals) as Δ_L:

$$\begin{aligned}\Delta_L := &\int_0^{\hat{v}(D)} \big(\theta + (1-\theta)(1-\varphi)\big)\lambda v f(v)\mathrm{d}v + \int_{\hat{v}(D)}^{\infty} \big(D + (1-\varphi)(v - D)\big) f(v)\mathrm{d}v \\ &- \int_0^{\hat{v}(D_0)} \lambda\theta v f(v)\mathrm{d}v - \int_{\hat{v}(D_0)}^{\infty} D_0 f(v)\mathrm{d}v.\end{aligned} \tag{90}$$

  Its IC is thus $\Delta_L \geq 0$.

- *Small creditors' IC.* Define their difference in payoffs from accepting and rejecting the offer as Δ_S:

$$\begin{aligned}\Delta_S := &\int_0^{\hat{v}(D)} (\theta + (1-\theta)(1-\varphi))\lambda v f(v)\mathrm{d}v \\ &+ \int_{\hat{v}(D)}^{\infty} (D + (1-\varphi)(v-D)) f(v)\mathrm{d}v - \int_{\hat{v}(D)}^{\infty} D_0 f(v)\mathrm{d}v.\end{aligned} \tag{91}$$

  Their IC is thus $\Delta_S \geq 0$.

Note the difference between these conditions: Whereas L takes into account the effect of restructuring on the default probability, S does not (this creates the hold-out problem in the baseline model).

When equity holders make a restructuring offer, they can choose D and $1-\varphi$ that will be accepted if there is a large creditor, if there are small creditors, or in both cases. Thus, they have to take into account which IC is tighter.

Comparing Δ_L and Δ_S reveals that L's IC is tighter than S's if

$$\Delta_S - \Delta_L = \int_0^{\hat{v}(D_0)} \lambda\theta v f(v)\mathrm{d}v - \int_{\hat{v}(D)}^{\hat{v}(D_0)} D_0 f(v)\mathrm{d}v \geq 0. \tag{92}$$

Because this inequality does not depend on φ, it is satisfied whenever D is above a threshold, $\tilde{D}$, which solves

$$\Delta_L - \Delta_S \Big|_{D=\tilde{D}} = 0. \tag{93}$$

Step 2: Current equity's payoff u for $\xi = 1$ and $\xi = 0$. To find the firm's payoff, we define its payoff in the following benchmark cases:

- u_L is the current equity holders' payoff if there is a large concentrated creditor (i.e. if $\xi = 1$):

$$u_L := \int_0^{\hat{v}(D)} \lambda v f(v)\mathrm{d}v + \int_{\hat{v}(D)}^{\infty} v f(v)\mathrm{d}v - \left(\int_0^{\hat{v}(D_0)} \lambda\theta v f(v)\mathrm{d}v + \int_{\hat{v}(D_0)}^{\infty} D_0 f(v)\mathrm{d}v \right). \tag{94}$$

- u_S is current equity's payoff if there are small dispersed creditors (i.e. if $\xi = 0$ as in the baseline model):

$$u_S := \int_0^{\hat{v}(D)} \lambda v f(v) \mathrm{d}v + \int_{\hat{v}(D)}^{\infty} v f(v) \mathrm{d}v - \int_{\hat{v}(D)}^{\infty} D_0 f(v) \mathrm{d}v. \tag{95}$$

- $u_\varnothing$ is current equity's payoff if there is no restructuring (i.e. if $D = D_0$):

$$u_\varnothing := \int_0^{\hat{v}(D_0)} \lambda(1-\theta) v f(v) \mathrm{d}v + \int_{\hat{v}(D_0)}^{\infty} (v - D_0) f(v) \mathrm{d}v. \tag{96}$$

These expressions are useful, because equity holders' payoff for $\xi \in (0,1)$ will be a weighted average of them.

Step 3: Current equity payoff calculation. There are three possible cases, depending on which IC binds, which we consider in turn.

Case 1: $\Delta_L \geq 0 > \Delta_S$. In this case, only L accepts the restructuring but S do not. So the restructuring succeeds with probability ξ and equity holders' problem is to

$$\text{maximize } u = \xi u_L + (1-\xi) u_\varnothing \tag{97}$$

over φ and D. Given $u_\varnothing$ does not depend on φ or D, this is the same as maximizing u_L. Differentiating with respect to D, we find

$$-(1-\lambda)\hat{v} f(\hat{v}) < 0, \tag{98}$$

for all D. So the solution is $D = 0$. We can find $1 - \varphi$ from the binding IC ($\Delta_L = 0$):

$$(1-\varphi)\mathbb{E}[v] = \int_0^{\hat{v}(D_0)} \lambda\theta v f(v) \mathrm{d}v + \int_{\hat{v}(D_0)}^{\infty} D_0 f(v) \mathrm{d}v. \tag{99}$$

Defining $u_L^{\max}$ as the maximum of u_L in this case, we have

$$u_L^{\max} = \int_0^\infty v f(v)\mathrm{d}v - \left(\int_0^{\hat{v}(D_0)} \lambda\theta v f(v)\mathrm{d}v + \int_{\hat{v}(D_0)}^\infty D_0 f(v)\mathrm{d}v \right). \tag{100}$$

So the expected payoff is

$$u = \xi u_L^{\max} + (1-\xi)u_\varnothing. \tag{101}$$

Case 2: $\Delta_S \geq 0 > \Delta_L$. In this case, only the small creditors' IC is satisfied. So the restructuring succeed with probability $1-\xi$ and and equity holders' problem is to

$$\text{maximize } u = \xi u_\varnothing + (1-\xi)u_S \tag{102}$$

over φ and D. Given $u_\varnothing$ does not depend on φ or D, this is the same as maximizing u_S. This is the baseline case of dispersed creditors studied in Section E: $1-\varphi = 0$ and $D = D^*$.

Defining $u_S^{\max}$ as the maximum of u_S in this case, we have

$$u_S^{\max} = \int_0^{\hat{v}(D)} \lambda v f(v)\mathrm{d}v + \int_{\hat{v}(D)}^\infty v f(v)\mathrm{d}v - \int_{\hat{v}(D)}^\infty D_0 f(v)\mathrm{d}v. \tag{103}$$

So the expected payoff is

$$u = \xi u_\varnothing + (1-\xi)u_S^{\max}. \tag{104}$$

Case 3: $\Delta_S \geq 0$ *and* $\Delta_L \geq 0$. In this case, both ICs are satisfied and equity holders' problem is to

$$\text{maximize } u = \xi u_L + (1-\xi)u_S \tag{105}$$

over φ and D.

Here, there are three sub-cases:

- Case 3(i): $\Delta_L > \Delta_S = 0$ ($\implies D < \tilde{D}$). In this case, the firm chooses φ and D so that S's IC binds and L accepts the restructuring on S's terms. Hence, the equity holders problem is to

$$\text{maximize } u = \xi u_S + (1-\xi)u_S = u_S \tag{106}$$

over φ and D. From case 2, we know that to maximize u_S, equity holders set $1-\varphi = 0$ and $D = D^*$. So the expected payoff is

$$u = u_S^{\max}. \tag{107}$$

- Case 3(ii): $\Delta_S > \Delta_L = 0$ ($\implies D > \tilde{D}$). This cannot arise in equilibrium: Given S's IC is slack ($\Delta_S > 0$), equity holders can increase their surplus by decreasing D (and increasing $1-\varphi$ so as not to violate L's IC).

- Case 3(iii): $\Delta_S = \Delta_L = 0$ ($\implies D = \tilde{D}$). Given both ICs are binding, $D = \tilde{D}$ by definition. Here, the expected payoff is

$$u = u_S(\tilde{D}) = u_L(\tilde{D}). \tag{108}$$

In this case, the offer includes both debt $\tilde{D}$ and an equity stake $1-\varphi$, which we can solve for from the binding ICs:

$$1-\varphi = \frac{\int_0^{\hat{v}_{D_0}} \lambda\theta v f(v)\mathrm{d}v + \int_{\hat{v}(D_0)}^{\infty} D_0 f(v)\mathrm{d}v - \int_0^{\hat{v}(\tilde{D})} \lambda\theta v f(v)\mathrm{d}v - \int_{\hat{v}(\tilde{D})}^{\infty} \tilde{D} f(v)\mathrm{d}v}{\int_0^{\hat{v}(\tilde{D})} (1-\theta)\lambda v f(v)\mathrm{d}v + \int_{\hat{v}(\tilde{D})}^{\infty} (v-\tilde{D}) f(v)\mathrm{d}v}. \tag{109}$$

Combining these sub-cases, we have that equity's expected payoff is

$$u = \begin{cases} u_S^{\max} & \text{if } D^* < \tilde{D}, \\ u_S(\tilde{D}) & \text{if } D^* \geq \tilde{D}. \end{cases} \tag{110}$$

Step 4: Payoff comparison. Combining the cases above, we have that equity's expected payoff is

$$u = \begin{cases} \max\left\{\xi u_L^{\max} + (1-\xi)u_\varnothing\,,\, \xi u_\varnothing + (1-\xi)u_S^{\max}\,,\, u_S^{\max}\right\} & \text{if } D^* < \tilde{D}, \\ \max\left\{\xi u_L^{\max} + (1-\xi)u_\varnothing\,,\, \xi u_\varnothing + (1-\xi)u_S^{\max}\,,\, u_S(\tilde{D})\right\} & \text{if } D^* \geq \tilde{D}. \end{cases} \tag{111}$$

To simplify the thresholds above, first observe that, from the definitions of $u_L^{\max}, u_S^{\max}$, and $u_S(\tilde{D})$ and the assumption that D_0 is sufficiently large (and hence $u_\varnothing$ sufficiently small), we have

$$u_L^{\max} \geq u_L(\tilde{D}) = u_S(\tilde{D}) > u_\varnothing \tag{112}$$

and

$$u_S^{\max} \geq u_L(\tilde{D}) = u_S(\tilde{D}) > u_\varnothing. \tag{113}$$

With this, we can rewrite u as

$$u = \begin{cases} \max\left\{\xi u_L^{\max} + (1-\xi)u_\varnothing\,,\, u_S^{\max}\right\} & \text{if } D^* < \tilde{D} \\ \max\left\{\xi u_L^{\max} + (1-\xi)u_\varnothing\,,\, \xi u_\varnothing + (1-\xi)u_S^{\max}\,,\, u_S(\tilde{D})\right\} & \text{if } D^* \geq \tilde{D}. \end{cases} \tag{114}$$

To give the formulation in the statement of the result, we divide the expression above into cases for $D^* \lesseqgtr \tilde{D}$ and $\underline{\xi} \lesseqgtr \bar{\xi}$, where

$$\underline{\xi} := \frac{u_S^{\max} - u_S(\tilde{D})}{u_S^{\max} - u_\varnothing}, \tag{115}$$

$$\bar{\xi} := \frac{u_S(\tilde{D}) - u_\varnothing}{u_L^{\max} - u_\varnothing}. \tag{116}$$

- If $D^* \geq \tilde{D}$ and $\underline{\xi} < \bar{\xi}$, then

$$u = \begin{cases} \xi u_\varnothing + (1-\xi) u_S^{\max} & \text{if } \xi \leq \underline{\xi}, \\ u_S(\tilde{D}) & \text{if } \xi \in \left(\underline{\xi}, \bar{\xi}\right], \\ \xi u_L^{\max} + (1-\xi) u_\varnothing & \text{if } \xi > \bar{\xi} \end{cases} \tag{117}$$

  The analysis above implies this corresponds to debt for $\xi \leq \underline{\xi}$, a mix of debt and equity for $\xi \in (\underline{\xi}, \bar{\xi}]$, and equity for $\xi > \bar{\xi}$.

- If $D^* \geq \tilde{D}$ and $\underline{\xi} \geq \bar{\xi}$, then

$$u = \begin{cases} \xi u_\varnothing + (1-\xi) u_S^{\max} & \text{if } \xi \leq \dfrac{u_S^{\max} - u_\varnothing}{u_L^{\max} + u_S^{\max} - 2u_\varnothing}, \\ \xi u_L^{\max} + (1-\xi) u_\varnothing & \text{if } \xi > \dfrac{u_S^{\max} - u_\varnothing}{u_L^{\max} + u_S^{\max} - 2u_\varnothing}. \end{cases} \tag{118}$$

  The analysis above implies this corresponds to debt for ξ below the threshold $(u_S^{\max} - u_\varnothing)/(u_L^{\max} + u_S^{\max} - 2u_\varnothing)$ and equity above it.

- If $D^* < \tilde{D}$, then

$$u = \begin{cases} u_S^{\max} & \text{if } \xi \leq \frac{u_S^{\max} - u_\varnothing}{u_L^{\max} - u_\varnothing} \\ \xi u_L^{\max} + (1-\xi) u_\varnothing & \text{if } \xi > \frac{u_S^{\max} - u_\varnothing}{u_L^{\max} - u_\varnothing}. \end{cases} \tag{119}$$

  The analysis above implies this corresponds to debt for ξ below the threshold $(u_S^{\max} - u_\varnothing)/(u_L^{\max} - u_\varnothing)$ and equity above it. □

F.15 Proof of Proposition 10

The proof is in the text. □

F.16 Proof of Proposition 11

We first rewrite the creditors' biding IC in inequality (4) as a function of η as

$$\Delta = \int_{\hat{v}(D)}^{\infty} \Big((1-\lambda^{\eta}(\hat{v})(1-\theta))\hat{v} - D_0 \Big) \mathrm{d}F^{\eta}(v) + \int_0^{\hat{v}(D)} \lambda^{\eta}(\hat{v})\theta v \mathrm{d}F^{\eta}(v) = 0, \tag{120}$$

having substituted for $D = (1-\lambda^{\eta}(\hat{v})(1-\theta))\hat{v}$ from equation (3). Solving, we can write the minimum $\lambda^{\eta}(\hat{v})$ needed to restructure to the debt level D as

$$\lambda^{\eta}(\hat{v}) = \frac{\int_{\hat{v}(D)}^{\infty}(D_0 - \hat{v})\mathrm{d}F^{\eta}(v)}{\int_0^{\hat{v}(D)} \theta v \mathrm{d}F^{\eta}(v) - \int_{\hat{v}}^{\infty}(1-\theta)\hat{v}\mathrm{d}F^{\eta}(v)} \tag{121}$$

$$= \frac{(D_0 - \hat{v})(1 - F^{\eta}(\hat{v}))}{\theta F^{\eta}(\hat{v})\mathbb{E}^{\eta}[v|v < \hat{v}] - (1-\theta)(1-F^{\eta}(\hat{v}))\hat{v}}, \tag{122}$$

where $\mathbb{E}^{\eta}$ denotes the expectation given the distribution F^{η}.

Now we show each part of the result in turn.

Tail expectation $\mathbb{E}^{\eta}[v|v < \hat{v}]$ does not depend on η. Given that, by assumption $F^{\eta'} \succ F^{\eta}$, we have that $F^{\eta'}(\hat{v}) < F^{\eta}(\hat{v})$. This implies that the numerator in equation (122) is increasing in η and the denominator is decreasing in η. Thus, $\lambda^{\eta'}(\hat{v}) > \lambda^{\eta}(\hat{v})$.

Default probability $F^{\eta}(\hat{v})$ does not depend on η. That is, $F^{\eta'}(\hat{v}) = F^{\eta}(\hat{v})$. Given that, by assumption $F^{\eta'} \succ F^{\eta}$, we have that $F^{\eta'}(\hat{v})\mathbb{E}^{\eta'}[v|v < \hat{v}] > F^{\eta}(\hat{v})\mathbb{E}^{\eta}[v|v < \hat{v}]$.[66] This implies

[66] Given $F^{\eta}(\hat{v})$ does not depend on η, this is akin to the fact that $F^{\eta'} \succ F^{\eta}$ implies $\mathbb{E}^{\eta'}[v] > \mathbb{E}^{\eta}[v]$ and can be proved via a change of variables, $v := (F^{\eta})^{-1}\big(F^{\eta'}(\tilde{v})\big)$:

$$\begin{aligned} F^{\eta}(\hat{v})\mathbb{E}^{\eta}[v|v < \hat{v}] \equiv \int_0^{\hat{v}} v \mathrm{d}F^{\eta}(v) &= \int_0^{\hat{v}} (F^{\eta})^{-1}\big(F^{\eta'}(\tilde{v})\big)\mathrm{d}F^{\eta}\Big((F^{\eta})^{-1}\big(F^{\eta'}(\tilde{v})\big)\Big) \\ &= \int_0^{\hat{v}} (F^{\eta})^{-1}\big(F^{\eta'}(\tilde{v})\big)\mathrm{d}F^{\eta'}(\tilde{v}) \\ &\leq \int_0^{\hat{v}} \tilde{v}\mathrm{d}F^{\eta'}(\tilde{v}) \equiv F^{\eta'}(\hat{v})\mathbb{E}^{\eta'}[v|v < \hat{v}], \end{aligned}$$

where the last inequality follows from the definition of stochastic dominance. That is, $F^{\eta}(\tilde{v}) \geq F^{\eta'}(\tilde{v})$ or, equivalently, $\tilde{v} \geq (F^{\eta})^{-1}\big(F^{\eta'}(\tilde{v})\big)$. Note, critically, that the assumption that $F^{\eta}(\hat{v})$ does not depend on η allowed that the denominator of equation (122) is increasing in η. And, since $F^{\eta'}(\hat{v}) = F^{\eta}(\hat{v})$, the numerator is the same under η and η'. Thus, $\lambda^{\eta'}(\hat{v}) < \lambda^{\eta}(\hat{v})$. □

G Omitted Derivations from Section 4.3

G.1 Derivation of Inequality (8)

The assumption that D^* is a continuous function of θ with a unique local maximum implies that a necessary and sufficient condition for higher θ to decrease the write-down is

$$\frac{\partial D^*}{\partial \theta} > 0. \tag{123}$$

Given equations (60) and (61), that is equivalent to saying

$$\begin{aligned} 0 > \frac{\partial \Delta}{\partial \theta} &= \lambda \int_0^{\hat{v}} v \mathrm{d}F(v) - \big(\lambda\theta\hat{v} + D_0 - D^*\big) f(\hat{v}) \frac{\partial \hat{v}}{\partial \theta} & (124) \\ &= \lambda \int_0^{\hat{v}} v \mathrm{d}F(v) - \big(\lambda\theta\hat{v} + D_0 - D^*\big) f(\hat{v}) \frac{\lambda \hat{v}}{1-(1-\theta)\lambda}. & (125) \end{aligned}$$

Now use the assumption that $vf(v)$ is increasing on $[0, \hat{v}]$ to see that

$$\int_0^{\hat{v}} v \mathrm{d}F(v) = \int_0^{\hat{v}} v f(v) \mathrm{d}v \leq \int_0^{\hat{v}} \max_{v\in[0,\hat{v}]} v f(v) \mathrm{d}v = \max_{v\in[0,\hat{v}]} v f(v) \int_0^{\hat{v}} \mathrm{d}v = \hat{v}^2 f(\hat{v}). \tag{126}$$

Using the above, we can get a sufficient condition for (126):

$$\hat{v}^2 f(\hat{v}) < \big(\lambda\theta\hat{v} + D_0 - D^*\big) f(\hat{v}) \hat{v} \frac{1}{1-(1-\theta)\lambda}. \tag{127}$$

Rearranging gives condition (8) in the text. □

us to change variables without changing the bounds of integration (the result would not obtain without that assumption).

G.2 Sources of Proxies for λ, θ, and D^*/D_0

The sources of the numbers usde for λ, θ, and D^*/D_0 in Section 4.3 are as follows:

- λ: This term captures the direct costs of bankruptcy. In studies of corporate reorganizations (mostly involving large corporations), the literature consistently estimates $\lambda > 90$ percent. (See Hotchkiss et al. (2008), Table 1, for a summary of twelve studies.)

- θ: A number of papers investigate the value retained by equity holders in bankruptcy. They suggest that $\theta > 85$ percent is a conservative lower bound. In most cases, creditors are paid in full before equity is paid anything. (See Hotchkiss et al. (2008), Section 5.1, for a summary of estimates.)

- D^*/D_0: As mentioned in the text, Mooradian and Ryan (2005) find a mean write-down of 44 percent.

H Internet Appendix: Further Results on Secured Creditor Power

Here we use the framework of Section 6.1 to ask how secured creditor control interacts with creditor friendliness and inefficient liquidation in bankruptcy.

H.1 Interaction of Secured Creditor Power and Creditor Friendliness

We now ask how secured creditor power affects the write-down-maximizing level of creditor friendliness: If secured creditors recover more relative to unsecured creditors (ρ is higher), should creditors as a whole get more or less relative to equity (a higher or lower θ) in order to maximize the write-down $D_0 - D$? We find that they should get more:

Proposition 12. *Suppose that the write-down is maximized at a unique interior level of creditor friendliness θ^* that is not an inflection point (as in, e.g., the uniform case in Figure 3). Increasing the secured creditor power ρ increases the optimal level of creditor friendliness. That is,* $\mathrm{d}\theta^*/\mathrm{d}\rho > 0$.

To see the intuition for this result, recall that θ^* is chosen to maximize the value of priority, balancing the increase in creditor recovery value against the decrease in the filing probability. Because high secured creditor power ρ increases recovery value without affecting the filing probability, θ^* increases to balance the two effects.

Proof. The IC in inequality (137) can be rewritten as

$$\Delta = (1 - F(\hat{v}))(D - D_0) + \left(1 - (1-\rho)\frac{D_0}{D}\right)\lambda\theta\int_0^{\hat{v}} v\mathrm{d}F(v) \geq 0. \tag{128}$$

The binding IC, $\Delta = 0$, defines the written-down debt level D^*; minimizing D^* over θ defines the optimal level of creditor friendliness. Thus, by the chain rule, the effect of ρ on θ^* is given by

$$\frac{\mathrm{d}\theta^*}{\mathrm{d}\rho} = -\frac{\frac{\partial}{\partial\rho}\left(\frac{\partial\Delta}{\partial\theta}\right)}{\frac{\partial}{\partial\theta}\left(\frac{\partial\Delta}{\partial\theta}\right)} = -\frac{\frac{\partial^2\Delta}{\partial\rho\partial\theta}}{\frac{\partial^2\Delta}{\partial\theta^2}}. \tag{129}$$

Note that the denominator is negative at θ^* given that we have assumed that θ^* is an interior local minimum and not an inflection point.[67]

We compute the denominator directly, step by step:

- First,

$$\begin{aligned}\frac{\partial\Delta}{\partial\theta} =& \left[f(\hat{v})(D_0 - D) + \left(1-(1-\rho)\frac{D_0}{D}\right)\lambda\theta\hat{v}f(\hat{v})\right]\frac{\partial\hat{v}}{\partial\theta} \\ &+ \left(1-(1-\rho)\frac{D_0}{D}\right)\lambda\int_0^{\hat{v}} v\mathrm{d}F(v) \end{aligned} \tag{130}$$

$$= -\lambda\hat{v}f(\hat{v})(\hat{v}_0 - \hat{v}) + \left(1-(1-\rho)\frac{D_0}{D}\right)\lambda\left(\int_0^{\hat{v}} v\mathrm{d}F(v) - \frac{\lambda\theta\hat{v}f(\hat{v})}{1-\lambda(1-\theta)}\hat{v}\right), \tag{131}$$

having used that $D = (1-\lambda(1-\theta))\hat{v}$, $D_0 = (1-\lambda(1-\theta))\hat{v}_0$, and

$$\frac{\partial\hat{v}}{\partial\theta} = -\frac{\lambda\hat{v}}{1-\lambda(1-\theta)}. \tag{132}$$

- Second,

$$\frac{\partial^2\Delta}{\partial\rho\partial\theta} = \frac{D_0}{D}\lambda\left(\int_0^{\hat{v}} v\mathrm{d}F(v) - \frac{\lambda\theta\hat{v}f(\hat{v})}{1-\lambda(1-\theta)}\hat{v}\right). \tag{133}$$

To determine its sign, it turns out that we can use two facts:

[67] To see why, differentiate the defining condition for θ^*, i.e. $\frac{\mathrm{d}D^*}{\mathrm{d}\theta}\Big|_{\theta=\theta^*} = -\frac{\partial\Delta/\partial\theta}{\partial\Delta/\partial D}\Big|_{\theta=\theta^*, D=D^*} = 0$, to get

$$\begin{aligned}\frac{\mathrm{d}^2 D^*}{\mathrm{d}\theta^2} &= -\left(\frac{\partial\Delta}{\partial D}\right)^{-2}\left[\frac{\partial\Delta}{\partial D}\frac{\mathrm{d}}{\mathrm{d}\theta}\left(\frac{\partial\Delta}{\partial\theta}\right) - \frac{\mathrm{d}}{\mathrm{d}\theta}\left(\frac{\partial\Delta}{\partial D}\right)\frac{\partial\Delta}{\partial\theta}\right] \\ &= -\left(\frac{\partial\Delta}{\partial D}\right)^{-1}\frac{\partial^2\Delta}{\partial\theta^2}\Big|_{\theta=\theta^*, D=D^*} > 0,\end{aligned}$$

having simplified using the conditions of optimality for θ^*, i.e. $\frac{\mathrm{d}D^*}{\mathrm{d}\theta}|_{\theta=\theta^*} = 0$ and $\frac{\partial\Delta}{\partial\theta}|_{\theta=\theta^*} = 0$. The result follows from the fact that $\frac{\partial\Delta}{\partial D} > 0$ (see, e.g., equation (58)).

– Using $D < D_0$ in $\Delta = 0$ above, we get that

$$1 - (1-\rho)\frac{D_0}{D} > 0. \tag{134}$$

– Given θ^* is optimal, $\partial\Delta/\partial\theta|_{\theta=\theta^*} = 0$. This, together with the last inequality, implies that

$$\int_0^{\hat{v}} v \mathrm{d}F(v) > \frac{\lambda\theta\hat{v}f(\hat{v})}{1-\lambda(1-\theta)}\hat{v}. \tag{135}$$

This implies that $\partial^2\Delta/\partial\rho\partial\theta$ in equation (133) is positive and, thus, given the above, that

$$\frac{\mathrm{d}\theta^*}{\mathrm{d}\rho} > 0. \tag{136}$$

□

H.2 Inefficiencies of Secured Creditor Control

Here we relax the assumption that liquidation costs do not depend on the division of surplus. We assume instead that secured creditor power can lead to inefficient liquidation. These costs could be borne by creditors or equity holders. We consider each case in turn.

H.2.1 Creditors Bear the Costs of Secured Creditor Power

First, we assume that unsecured debt receives only a fraction ζ of what is left after secured debt and equity are paid, so $1-\zeta$ captures the inefficiency of secured creditor power. With this modification, the creditors' IC in equation (**??**) becomes

$$\begin{aligned}\Big(1 - F\big(\hat{v}(D)\big)\Big)D + F\big(\hat{v}(D)\big)\mathbb{E}\left[\lambda\theta v \,\middle|\, v < \hat{v}(D)\right] \\ \geq \Big(1 - F\big(\hat{v}(D)\big)\Big)D_0 + \zeta(1-\rho)F\big(\hat{v}(D)\big)\mathbb{E}\left[\lambda\theta v \,\middle|\, v < \hat{v}(D)\right]\frac{D_0}{D}.\end{aligned} \tag{137}$$

Observe that ζ above plays the same role that $1-\rho$ does (cf. equation (17)). Therefore, Proposition **??** and Proposition 12 imply that an increase in ζ makes restructuring harder and reduces the optimal level of creditor friendliness θ^*.

H.2.2 Equity Holders Bear the Cost of Secured Creditor Power

Now we assume that for creditors to gain \$1 in bankruptcy, equity holders must forgo more than \$1. Specifically, for every $(1-\gamma/2)\lambda\theta v$ that creditors get, equity gives up $(1-\theta)\lambda v$. Thus, γ measures the inefficiencies they induce ex post. If $\gamma = 0$, the model is the same as the baseline. Increasing γ decreases the total surplus.

To explore how the inefficiencies of creditor power could affect restructuring, we explore how the optimal level of creditor friendliness depends on the inefficiencies induced by secured creditor power γ (cf. Section 4.2 and Section 4.3). This gives the next result:

Proposition 13. *Suppose that the write-down is maximized at a unique interior level of creditor friendliness θ^* that is not an inflection point (as in, e.g., the uniform case in Figure 3). Increasing the inefficiency of secured creditor power γ decreases the optimal level of creditor friendliness. That is,* $\mathrm{d}\theta^*/\mathrm{d}\gamma < 0$.

Intuitively, if giving creditors power destroys value in bankruptcy, then giving them more power is likely to make them even less willing to accept a restructuring. Hence, the larger is γ, the larger is the region of θ for which making the code more creditor friendly makes restructuring harder.

Proof. We begin from the creditors' IC:

$$\Delta = (1-F(\hat{v}))(D-D_0) + \lambda\theta\left(1-\frac{\gamma\theta}{2}\right)\int_0^{\hat{v}} v\mathrm{d}F(v), \tag{138}$$

which is just the creditors' IC in equation (4) modified to include the inefficiencies captured by γ . The binding IC, $\Delta = 0$, defines the written-down debt level D^* and minimizing D^* over θ defines the optimal level of creditor friendliness θ^*. Thus, by the chain rule, the effect of γ on θ^*

is given by:

$$\frac{\mathrm{d}\theta^*}{\mathrm{d}\gamma} = -\frac{\frac{\partial}{\partial\gamma}\left(\frac{\partial\Delta}{\partial\theta}\right)}{\frac{\partial}{\partial\theta}\left(\frac{\partial\Delta}{\partial\theta}\right)} = -\frac{\partial^2\Delta/(\partial\gamma\partial\theta)}{\partial^2\Delta/\partial\theta^2}. \tag{139}$$

Note that the denominator is negative given that we have assumed that θ^* is an interior local minimum and not an inflection point (see footnote 67).

We compute the numerator $\frac{\partial^2\Delta}{\partial\gamma\partial\theta}$ directly, step by step:

- First,

$$\begin{aligned}\frac{\partial\Delta}{\partial\theta} =& \lambda\left(1-\gamma\theta\right)\int_0^{\hat{v}} v\mathrm{d}F(v) + \left[D_0 - D + \lambda\theta\left(1-\frac{\gamma\theta}{2}\right)\hat{v}\right] f(\hat{v})\frac{\partial\hat{v}}{\partial\theta} &(140)\\ =& \lambda\left(1-\gamma\theta\right)\int_0^{\hat{v}} v\mathrm{d}F(v) - \lambda\left[\hat{v}_0 - \hat{v} + \frac{\lambda\theta\left(2-\gamma\theta\right)}{2(1-\lambda(1-\theta))}\hat{v}\right]\hat{v}f(\hat{v}), &(141)\end{aligned}$$

  having used that $D = (1-\lambda(1-\theta))\hat{v}$, $D_0 = (1-\lambda(1-\theta))\hat{v}_0$, and

$$\frac{\partial\hat{v}}{\partial\theta} = \frac{-\lambda\hat{v}}{1-\lambda(1-\theta)}. \tag{142}$$

- Second,

$$\frac{\partial^2\Delta}{\partial\gamma\partial\theta} = -\lambda\theta\left(\int_0^{\hat{v}} v\mathrm{d}F(v) - \frac{\lambda\theta}{2(1-\lambda(1-\theta))}\hat{v}^2 f(\hat{v})\right). \tag{143}$$

To determine the sign, it turns out that we can use the fact that θ^* is optimal, so $\partial\Delta/\partial\theta|_{\theta=\theta^*} = 0$. That is, the derivative in equation (141) is zero. Manipulating, we get

$$\lambda\left(1-\gamma\theta\right)\left(\int_0^{\hat{v}} v\mathrm{d}F(v) - \frac{\lambda\theta}{2(1-\lambda(1-\theta))}\hat{v}^2 f(\hat{v})\right) - \lambda\left[\hat{v}_0 - \hat{v} + \frac{\lambda\theta}{2(1-\lambda(1-\theta))}\hat{v}\right]\hat{v}f(\hat{v}) = 0 \tag{144}$$

and, thus, that

$$-\int_0^{\hat{v}} v\mathrm{d}F(v) + \frac{\lambda\theta}{2(1-\lambda(1-\theta))}\hat{v}^2 f(\hat{v}) = -\left(1-\gamma\theta\right)^{-1}\left[\hat{v}_0 - \hat{v} + \frac{\lambda\theta}{2(1-\lambda(1-\theta))}\hat{v}\right]\hat{v}f(\hat{v}), \tag{145}$$

which is the numerator in equation (143). It is negative given that $\hat{v}_0 > \hat{v}$.

Substituting into equation (139), we see that $\mathrm{d}\theta^*/\mathrm{d}\gamma$ is negative.

$\square$